Supplementary Material

S-box implementations

The value **A** represents the number of AND gates and the value **X** the number of XOR gates.

5-bit S-box implementations

χ_5
A= 5 X= 10

$$
\begin{aligned}
l_0 &= x_4 \oplus x_2;\\
l_1 &= x_4 \oplus x_1;\\
l_2 &= x_3 \oplus x_1;\\
l_3 &= x_3 \oplus x_0;\\
l_4 &= x_2 \oplus x_0;\\
q_0 &= (x_2 \times x_1);\\
y_0 &= q_0 \oplus l_4;\\
q_1 &= (x_4 \times x_3);\\
y_2 &= q_1 \oplus l_0;\\
q_2 &= (x_3 \times x_2);\\
y_1 &= q_2 \oplus l_2;\\
q_3 &= (x_4 \times x_0);\\
y_3 &= q_3 \oplus l_3;\\
q_4 &= (x_1 \times x_0);\\
y_4 &= q_4 \oplus l_1;
\end{aligned}
$$

ASCON
A= 5 X= 15

$$
\begin{aligned}
l_0 &= x_4 \oplus x_3;\\
l_1 &= x_3 \oplus x_2;\\
l_2 &= x_3 \oplus x_1;\\
l_3 &= x_4 \oplus x_0;\\
l_4 &= x_2 \oplus l_3;\\
l_5 &= x_4 \oplus l_2;\\
l_6 &= l_1 \oplus l_5;\\
l_7 &= l_2 \oplus l_4;\\
l_8 &= x_4 \oplus l_7;\\
q_0 &= x_1 \times l_4;\\
y_0 &= q_0 \oplus l_8;\\
q_1 &= (l_1 \times l_2) \oplus x_3;\\
y_1 &= q_1 \oplus l_7;\\
q_2 &= x_1 \times l_3;\\
y_4 &= q_2 \oplus l_5;\\
q_3 &= x_0 \times l_0;\\
y_3 &= q_3 \oplus l_7;\\
q_4 &= x_4 \times x_3;\\
y_2 &= q_4 \oplus l_6 \oplus 1;
\end{aligned}
$$

FIDES
A= 7 X= 20

$$
\begin{aligned}
l_0 &= x_4 \oplus x_3;\\
l_1 &= x_4 \oplus x_2;\\
l_2 &= x_4 \oplus x_1;\\
l_3 &= x_3 \oplus x_1;\\
l_4 &= x_2 \oplus x_1;\\
l_5 &= x_3 \oplus x_0;\\
l_6 &= x_2 \oplus x_0;\\
l_7 &= x_1 \oplus x_0;\\
l_8 &= x_3 \oplus l_4;\\
q_0 &= x_3 \times l_4;\\
q_1 &= x_1 \times l_1;\\
q_2 &= l_1 \times l_3;\\
q_3 &= x_4 \times l_5;\\
q_4 &= l_0 \times l_6;\\
q_5 &= x_2 \times l_7;\\
q_6 &= x_0 \times l_3;\\
y_0 &= q_1 \oplus q_5 \oplus l_5 \oplus 1;\\
y_1 &= q_4 \oplus q_6 \oplus x_4;\\
y_2 &= q_3 \oplus q_5 \oplus l_0;\\
y_3 &= q_0 \oplus q_5 \oplus l_2;\\
y_4 &= q_2 \oplus q_3 \oplus q_4 \oplus l_8;
\end{aligned}
$$

X^3
A= 7 X= 19

$$
\begin{aligned}
l_0 &= x_4 \oplus x_3;\\
l_1 &= x_3 \oplus x_2;\\
l_2 &= x_3 \oplus x_1;\\
l_3 &= x_2 \oplus x_0;\\
l_4 &= x_4 \oplus l_3;\\
l_5 &= l_0 \oplus x_0;\\
l_6 &= x_4 \oplus x_2;\\
l_7 &= l_6 \oplus l_2;\\
q_0 &= x_2 \times l_0;\\
q_1 &= x_1 \times l_0;\\
q_2 &= (l_1 \times l_2) \oplus x_3;\\
q_3 &= x_4 \times x_0;\\
q_4 &= x_3 \times l_4;\\
q_5 &= l_2 \times l_5;\\
q_6 &= (l_3 \times l_7) \oplus x_2;\\
y_0 &= q2 \oplus q3 \oplus x_0;\\
y_1 &= q1 \oplus q5 \oplus l_6;\\
y_2 &= q4 \oplus q6 \oplus x_4;\\
y_3 &= q0 \oplus q3 \oplus l_7;\\
y_4 &= q5 \oplus q6;
\end{aligned}
$$

X^5
A= 7 X= 21

$l_0 = x_4 \oplus x_3;$
$l_1 = x_4 \oplus x_2;$
$l_2 = x_3 \oplus x_2;$
$l_3 = x_4 \oplus x_0;$
$l_4 = l_1 \oplus x_1;$
$l_5 = x_2 \oplus l_3;$
$l_6 = l_4 \oplus l_3;$
$l_7 = x_3 \oplus l_6;$
$l_8 = l_0 \oplus x_1;$
$q_0 = (x_4 \times l_2);$
$q_1 = (x_3 \times l_1);$
$q_2 = (l_0 \times l_4) \oplus x_4;$
$q_3 = (x_3 \times l_6);$
$q_4 = (x_0 \times l_1);$
$q_5 = (x_1 \times l_5);$
$q_6 = (l_3 \times l_8) \oplus x_4;$
$y_0 = q_0 \oplus q_4 \oplus l_7;$
$y_1 = q_1 \oplus q_6 \oplus x_3;$
$y_2 = q_2 \oplus q_3 \oplus l_8;$
$y_3 = q_2 \oplus q_4 \oplus l_0;$
$y_4 = q_1 \oplus q_5 \oplus l_2;$

S_51
A= 5 X= 20

$l_0 = x_4 \oplus x_3;$
$l_1 = x_4 \oplus x_2;$
$l_2 = x_3 \oplus x_2;$
$l_3 = x_3 \oplus x_1;$
$l_4 = x_2 \oplus x_1;$
$l_5 = x_2 \oplus x_0;$
$l_6 = x_1 \oplus x_0;$
$l_7 = x_4 \oplus l_4;$
$l_8 = l_0 \oplus x_0;$
$l_9 = x_4 \oplus l_2;$
$q_0 = x_1 \times l_9;$
$y_0 = q_0 \oplus l_6;$
$q_1 = l_4 \times l_8;$
$l_{10} = q_0 \oplus q_1;$
$q_2 = (l_6 \times l_7) \oplus x_1;$
$y_1 = q_1 \oplus q_2 \oplus l_4;$
$q_3 = x_2 \times l_3;$
$y_3 = q_3 \oplus l_{10} \oplus l_2;$
$y_2 = l_{10} \oplus l_1;$
$q_4 = x_3 \times l_5;$
$y_4 = q_4 \oplus q_1 \oplus l_0;$

S_52
A= 5 X= 24

$l_0 = x_4 \oplus x_3;$
$l_1 = x_4 \oplus x_2;$
$l_2 = x_4 \oplus x_1;$
$l_3 = x_3 \oplus x_1;$
$l_4 = x_4 \oplus x_0;$
$l_5 = x_2 \oplus x_0;$
$l_6 = x_3 \oplus l_1;$
$l_7 = x_4 \oplus l_3;$
$l_8 = x_2 \oplus l_2;$
$l_9 = x_3 \oplus l_4;$
$l_{10} = x_3 \oplus l_8;$
$q_0 = (l_4 \times l_6) \oplus x_4;$
$y_1 = q_0 \oplus l_2;$
$q_1 = (l_1 \times l_7) \oplus x_4;$
$y_3 = q_1 \oplus l_0;$
$l_{11} = q_0 \oplus q_1;$
$q_2 = (x_0 \times l_8);$
$y_4 = l_{11} \oplus q_2 \oplus l_4;$
$q_3 = (l_3 \times l_9) \oplus x_3;$
$y_2 = q_1 \oplus q_3 \oplus q_2 \oplus l_5;$
$q_4 = (x_0 \times l_{10});$
$y_0 = l_{11} \oplus q_4 \oplus x_0;$

S_53
A= 5 X= 24

$l_0 = x_4 \oplus x_3;$
$l_1 = x_4 \oplus x_2;$
$l_2 = x_3 \oplus x_2;$
$l_3 = x_3 \oplus x_1;$
$l_4 = x_4 \oplus x_0;$
$l_5 = x_3 \oplus x_0;$
$l_6 = x_2 \oplus x_0;$
$l_7 = l_1 \oplus l_3;$
$l_8 = x_1 \oplus l_4;$
$l_9 = x_4 \oplus l_7;$
$l_{10} = x_4 \oplus l_3;$
$q_0 = (l_9 \times l_8) \oplus x_1;$
$y_2 = q_0 \oplus l_1;$
$q_1 = (l_5 \times l_7) \oplus x_3;$
$y_3 = q_1 \oplus l_2;$
$q_2 = (x_3 \times l_1);$
$l_{11} = q_1 \oplus q_2;$
$y_0 = l_{11} \oplus l_4;$
$q_3 = (l_2 \times l_{10}) \oplus x_3;$
$y_4 = q_3 \oplus q_1 \oplus l_0;$
$q_4 = (l_3 \times l_6);$
$y_1 = l_{11} \oplus q_4 \oplus q_0 \oplus q_3 \oplus l_3;$

S_54
A = 6 X = 20

$$
\begin{aligned}
l_0 &= x_4 \oplus x_3;\\
l_1 &= x_3 \oplus x_2;\\
l_2 &= x_4 \oplus x_1;\\
l_3 &= x_3 \oplus x_1;\\
l_4 &= x_4 \oplus x_0;\\
l_5 &= x_2 \oplus x_0;\\
l_6 &= x_1 \oplus x_0;\\
l_7 &= x_4 \oplus l_1;\\
l_8 &= x_4 \oplus l_6;\\
q_0 &= (x_3 \times x_1);\\
q_1 &= (l_6 \times l_7);\\
y_0 &= q_0 \oplus q_1 \oplus l_4;\\
q_2 &= (x_3 \times l_8);\\
y_2 &= q_2 \oplus q_1 \oplus l_5;\\
q_3 &= (x_3 \times l_2);\\
l_9 &= q_1 \oplus q_3;\\
y_1 &= l_9 \oplus x_1;\\
q_4 &= (x_4 \times l_1);\\
q_5 &= (l_2 \times l_5);\\
l_{10} &= q_4 \oplus q_5;\\
y_4 &= l_{10} \oplus l_0;\\
y_3 &= l_{10} \oplus l_9 \oplus q_0 \oplus l_3;
\end{aligned}
$$

S-box	LUT
χ_5	0,9,18,11,5,12,22,15,10,3,24,1,13,4,30,7,20,21,6,23 17,16,2,19,26,27,8,25,29,28,14,31
ASCON	4,15,27,1,11,0,23,13,31,28,2,16,18,17,12,30,26,25 20,6,21,22,24,10,5,14,9,19,8,3,7,29
FIDES	1,0,25,26,17,29,21,27,20,5,4,23,14,18,2,28,15,8,6,3 13,7,24,16,30,9,31,10,22,12,11,19
X^3	0,1,8,15,10,31,23,4,26,25,3,6,9,30,5,20,14,18,22,12 24,16,21,27,2,28,11,19,13,7,17,29
X^5	0,1,5,22,17,25,4,30,31,24,18,7,20,26,9,21,12,6,23 15,16,19,27,10,14,2,29,3,8,13,11,28
S_51	0,1,3,30,14,19,10,11,24,9,8,5,16,29,7,22,20,23,6,25 4,27,17,18,12,31,13,2,26,21,28,15
S_52	0,21,2,6,4,23,27,25,8,11,14,28,17,20,10,30,26,9,1,3 16,5,22,18,24,29,7,19,15,12,13,31
S_53	0,1,2,28,12,16,24,27,26,4,23,22,30,29,5,25,21,15,17 20,13,10,31,7,3,6,8,18,19,11,14,9
S_54	0,5,10,23,4,14,25,11,24,22,26,12,28,29,9,16 17,3,20,30,21,8,7,2,31,6,18,19,27,13,1,15

6-bit S-box implementations

χ_6
A= 6 X= 12

$$
\begin{aligned}
l_0 &= x_5 \oplus x_3;\\
l_1 &= x_4 \oplus x_2;\\
l_2 &= x_5 \oplus x_1;\\
l_3 &= x_3 \oplus x_1;\\
l_4 &= x_4 \oplus x_0;\\
l_5 &= x_2 \oplus x_0;\\
q_0 &= (x_2 \times x_1);\\
y_0 &= q_0 \oplus l_5;\\
q_1 &= (x_5 \times x_4);\\
y_3 &= q_1 \oplus l_0;\\
q_2 &= (x_4 \times x_3);\\
y_2 &= q_2 \oplus l_1;\\
q_3 &= (x_3 \times x_2);\\
y_1 &= q_3 \oplus l_3;\\
q_4 &= (x_5 \times x_0);\\
y_4 &= q_4 \oplus l_4;\\
q_5 &= (x_1 \times x_0);\\
y_5 &= q_5 \oplus l_2;
\end{aligned}
$$

S_2256
A= 8 X= 26

$$
\begin{aligned}
l_0 &= x_5 \oplus x_3;\\
l_1 &= x_3 \oplus x_2;\\
l_2 &= x_5 \oplus x_1;\\
l_3 &= x_4 \oplus x_0;\\
l_4 &= x_2 \oplus x_0;\\
l_5 &= x_2 \oplus l_2;\\
l_6 &= l_2 \oplus x_0;\\
l_7 &= x_4 \oplus l_5;\\
l_8 &= x_3 \oplus l_6;\\
l_9 &= x_5 \oplus l_1;\\
l_{10} &= l_0 \oplus l_3;\\
l_{11} &= x_4 \oplus l_0;\\
q_0 &= (x_4 \times l_5);\\
q_1 &= (x_3 \times l_7);\\
q_2 &= (l_0 \times l_2) \oplus x_5;\\
q_3 &= (x_4 \times l_8);\\
q_4 &= (l_0 \times l_4);\\
q_5 &= (l_1 \times l_{10}) \oplus x_3;\\
q_6 &= (l_1 \times l_6);\\
q_7 &= (x_0 \times l_9);\\
l_{12} &= q_0 \oplus q_4;\\
l_{13} &= q_2 \oplus q_3;\\
y_0 &= q_0 \oplus l_{13} \oplus q_6 \oplus l_3;\\
y_1 &= l_{12} \oplus q_2 \oplus q_5 \oplus l_2;\\
y_2 &= l_{12} \oplus x_2;\\
y_3 &= q_1 \oplus q_3 \oplus l_{11};\\
y_4 &= l_{13};\\
y_5 &= q_5 \oplus q_7;
\end{aligned}
$$

S_2257
A= 8 X= 27

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_4 \oplus x_3;\\
l_2 &= x_5 \oplus x_1;\\
l_3 &= x_5 \oplus x_0;\\
l_4 &= x_4 \oplus x_0;\\
l_5 &= x_1 \oplus x_0;\\
l_6 &= l_1 \oplus x_2;\\
l_7 &= x_2 \oplus l_5;\\
l_8 &= x_4 \oplus l_6;\\
l_9 &= l_8 \oplus x_1;\\
l_{10} &= x_2 \oplus l_3;\\
l_{11} &= x_5 \oplus l_1;\\
q_0 &= (x_1 \times l_1);\\
q_1 &= (l_{11} \times l_9) \oplus x_3;\\
q_2 &= (x_2 \times x_1);\\
q_3 &= (x_4 \times l_{10});\\
q_4 &= (l_0 \times l_3) \oplus x_5;\\
q_5 &= (l_0 \times l_5);\\
q_6 &= (l_{11} \times l_{10}) \oplus x_5;\\
q_7 &= (l_6 \times l_7) \oplus x_2;\\
y_0 &= q_1 \oplus q_7 \oplus l_4;\\
y_1 &= q_2 \oplus q_6 \oplus q_7 \oplus l_2;\\
y_2 &= q_5 \oplus q_6 \oplus x_2;\\
y_3 &= q_1 \oplus q_3 \oplus l_{11};\\
y_4 &= q_0 \oplus q_4;\\
y_5 &= q_0 \oplus q_1;
\end{aligned}
$$

S_2258
A= 8 X= 25

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_5 \oplus x_3;\\
l_2 &= x_4 \oplus x_3;\\
l_3 &= x_5 \oplus x_1;\\
l_4 &= x_4 \oplus x_0;\\
l_5 &= x_2 \oplus x_0;\\
l_6 &= l_3 \oplus x_0;\\
l_7 &= l_2 \oplus x_2;\\
l_8 &= x_3 \oplus l_6;\\
l_9 &= x_5 \oplus l_2;\\
l_{10} &= x_3 \oplus l_5;\\
l_{11} &= x_5 \oplus l_8;\\
q_0 &= (x_1 \times l_1);\\
q_1 &= (x_1 \times l_2);\\
q_2 &= (x_4 \times l_{10});\\
q_3 &= (l_0 \times l_8) \oplus x_5;\\
q_4 &= (l_5 \times l_9);\\
q_5 &= (l_9 \times l_{11}) \oplus x_3;\\
q_6 &= (x_0 \times l_7);\\
q_7 &= (l_7 \times l_6);\\
l_{12} &= q_4 \oplus q_5;\\
y_0 &= q_1 \oplus q_7 \oplus l_4;\\
y_1 &= q_5 \oplus q_6 \oplus l_3;\\
y_2 &= q_3 \oplus q_4 \oplus x_2;\\
y_3 &= q_2 \oplus l_{12} \oplus l_9;\\
y_4 &= q_0 \oplus q_3;\\
y_5 &= q_1 \oplus l_{12};
\end{aligned}
$$

S_2259
A = 8 X = 28

$$l_0 = x_3 \oplus x_2;$$
$$l_1 = x_5 \oplus x_1;$$
$$l_2 = x_4 \oplus x_1;$$
$$l_3 = x_2 \oplus x_1;$$
$$l_4 = x_4 \oplus x_0;$$
$$l_5 = x_2 \oplus x_0;$$
$$l_6 = x_4 \oplus l_5;$$
$$l_7 = l_0 \oplus l_1;$$
$$l_8 = l_2 \oplus l_7;$$
$$l_9 = x_3 \oplus l_4;$$
$$l_{10} = x_5 \oplus l_9;$$
$$l_{11} = x_3 \oplus l_2;$$
$$l_{12} = x_3 \oplus l_5;$$
$$q_0 = (l_0 \times l_{11}) \oplus x_3;$$
$$q_1 = (x_2 \times l_9);$$
$$q_2 = (l_1 \times l_4);$$
$$q_3 = (l_1 \times l_{12});$$
$$q_4 = (l_2 \times l_{10}) \oplus x_4;$$
$$q_5 = (l_{11} \times l_6);$$
$$q_6 = (l_3 \times l_{10});$$
$$q_7 = (l_5 \times l_7) \oplus x_2;$$
$$y_0 = q_3 \oplus q_7 \oplus l_9;$$
$$y_1 = q_5 \oplus q_6 \oplus x_1;$$
$$y_2 = q_6 \oplus q_7 \oplus l_8;$$
$$y_3 = q_1 \oplus q_2 \oplus q_7 \oplus x_5;$$
$$y_4 = q_2 \oplus q_4;$$
$$y_5 = q_0 \oplus q_2 \oplus q_3;$$

S_2260
A= 8 X= 29

$$l_0 = x_5 \oplus x_4;$$
$$l_1 = x_5 \oplus x_3;$$
$$l_2 = x_4 \oplus x_2;$$
$$l_3 = x_3 \oplus x_2;$$
$$l_4 = x_4 \oplus x_1;$$
$$l_5 = x_3 \oplus x_0;$$
$$l_6 = x_1 \oplus x_0;$$
$$l_7 = x_5 \oplus l_3;$$
$$l_8 = l_7 \oplus l_5;$$
$$l_9 = l_1 \oplus l_4;$$
$$l_{10} = x_4 \oplus l_7;$$
$$l_{11} = l_3 \oplus x_1;$$
$$l_{12} = x_5 \oplus l_4;$$
$$l_{13} = x_4 \oplus l_1;$$
$$l_{14} = x_3 \oplus l_8;$$
$$q_0 = (l_1 \times l_4);$$
$$q_1 = (l_{13} \times l_{11}) \oplus x_3;$$
$$q_2 = (l_2 \times l_9);$$
$$q_3 = (x_5 \times l_5);$$
$$q_4 = (l_0 \times l_{14}) \oplus x_5;$$
$$q_5 = (l_{13} \times l_8) \oplus x_5;$$
$$q_6 = (l_3 \times l_6);$$
$$q_7 = (l_6 \times l_7);$$
$$l_{15} = q_2 \oplus q_5;$$
$$y_0 = q_2 \oplus q_6 \oplus l_5;$$
$$y_1 = q_4 \oplus q_7 \oplus l_{12};$$
$$y_2 = q_0 \oplus l_{15} \oplus l_{10};$$
$$y_3 = q_3 \oplus l_{15} \oplus x_5;$$
$$y_4 = q_0 \oplus q_4;$$
$$y_5 = q_0 \oplus q_1;$$

S_2261
A= 8 X= 26

$$l_0 = x_5 \oplus x_3;$$
$$l_1 = x_4 \oplus x_2;$$
$$l_2 = x_3 \oplus x_2;$$
$$l_3 = x_4 \oplus x_1;$$
$$l_4 = x_5 \oplus x_0;$$
$$l_5 = x_1 \oplus x_0;$$
$$l_6 = l_3 \oplus l_4;$$
$$l_7 = x_3 \oplus l_4;$$
$$l_8 = l_1 \oplus x_0;$$
$$l_9 = x_4 \oplus l_7;$$
$$l_{10} = l_8 \oplus l_6;$$
$$q_0 = (x_5 \times x_1);$$
$$q_1 = (l_0 \times l_{10}) \oplus x_5;$$
$$q_2 = (l_1 \times l_3);$$
$$q_3 = (x_4 \times l_5);$$
$$q_4 = (l_0 \times l_5);$$
$$q_5 = (l_2 \times l_4);$$
$$q_6 = (l_2 \times l_6);$$
$$q_7 = (l_7 \times l_8) \oplus x_0;$$
$$l_{11} = q_5 \oplus q_7;$$
$$y_0 = q_1 \oplus q_5 \oplus l_9;$$
$$y_1 = q_0 \oplus q_6 \oplus l_3;$$
$$y_2 = q_2 \oplus q_3 \oplus q_4 \oplus l_2;$$
$$y_3 = q_4 \oplus q_5 \oplus q_6 \oplus x_5;$$
$$y_4 = q_1 \oplus q_3 \oplus l_{11};$$
$$y_5 = l_{11};$$

S_2262
A= 8 X= 27

$$l_0 = x_4 \oplus x_3;$$
$$l_1 = x_4 \oplus x_2;$$
$$l_2 = x_4 \oplus x_1;$$
$$l_3 = x_4 \oplus x_0;$$
$$l_4 = x_3 \oplus l_1;$$
$$l_5 = x_5 \oplus x_3;$$
$$l_6 = l_5 \oplus l_3;$$
$$l_7 = x_5 \oplus l_1;$$
$$l_8 = x_4 \oplus l_6;$$
$$l_9 = x_5 \oplus l_2;$$
$$l_{10} = l_4 \oplus l_2;$$
$$l_{11} = l_9 \oplus l_6;$$
$$q_0 = (l_0 \times l_9) \oplus x_4;$$
$$q_1 = (l_2 \times l_7) \oplus x_4;$$
$$q_2 = (l_9 \times l_{10}) \oplus x_1;$$
$$q_3 = (x_2 \times l_3);$$
$$q_4 = (x_2 \times l_{11});$$
$$q_5 = (l_1 \times l_{11});$$
$$q_6 = (x_0 \times l_7);$$
$$q_7 = (l_4 \times l_8);$$
$$l_{12} = q_2 \oplus q_6;$$
$$y_0 = q_2 \oplus q_4 \oplus l_6;$$
$$y_1 = l_{12} \oplus l_2;$$
$$y_2 = q_0 \oplus q_1 \oplus l_4;$$
$$y_3 = q_5 \oplus q_7 \oplus l_5;$$
$$y_4 = q_1 \oplus q_3 \oplus l_{12};$$
$$y_5 = q_1 \oplus q_5 \oplus q_6;$$

S_2263
A= 8 X= 30

$l_0 = x_5 \oplus x_4;$
$l_1 = x_4 \oplus x_2;$
$l_2 = x_3 \oplus x_2;$
$l_3 = x_5 \oplus x_1;$
$l_4 = x_4 \oplus x_1;$
$l_5 = x_5 \oplus x_0;$
$l_6 = x_4 \oplus l_2;$
$l_7 = x_1 \oplus l_5;$
$l_8 = x_5 \oplus l_1;$
$l_9 = x_2 \oplus l_3;$
$l_{10} = x_3 \oplus l_4;$
$l_{11} = x_3 \oplus l_7;$
$l_{12} = x_3 \oplus l_5;$
$l_{13} = l_9 \oplus l_{12};$
$q_0 = (x_1 \times l_0);$
$q_1 = (l_2 \times l_3);$
$q_2 = (l_{10} \times l_9) \oplus x_1;$
$q_3 = (x_5 \times l_{13});$
$q_4 = (x_2 \times l_5);$
$q_5 = (l_1 \times l_{11});$
$q_6 = (l_8 \times l_7) \oplus x_5;$
$q_7 = (l_6 \times l_{12}) \oplus x_3;$
$l_{14} = q_2 \oplus q_3;$
$y_0 = q_0 \oplus l_{14} \oplus q_6 \oplus l_{12};$
$y_1 = l_{14} \oplus l_4;$
$y_2 = q_1 \oplus q_3 \oplus q_4 \oplus l_6;$
$y_3 = q_3 \oplus q_7 \oplus x_5;$
$y_4 = q_4 \oplus q_6;$
$y_5 = q_2 \oplus q_4 \oplus q_5;$

X^3
A= 8 X= 37

$l_0 = x_5 \oplus x_4;$
$l_1 = x_5 \oplus x_3;$
$l_2 = x_3 \oplus x_2;$
$l_3 = x_5 \oplus x_1;$
$l_4 = x_1 \oplus x_0;$
$l_5 = l_1 \oplus x_0;$
$l_6 = x_4 \oplus l_1;$
$l_7 = l_2 \oplus l_5;$
$l_8 = x_5 \oplus l_4;$
$l_9 = l_6 \oplus l_7;$
$l_{10} = l_9 \oplus l_4;$
$l_{11} = x_3 \oplus l_3;$
$l_{12} = x_4 \oplus l_4;$
$l_{13} = x_4 \oplus l_3;$
$l_{14} = l_{11} \oplus l_9;$
$l_{15} = x_3 \oplus l_{10};$
$q_0 = (x_5 \times l_2);$
$q_1 = (l_0 \times l_{11}) \oplus x_5;$
$q_2 = (x_3 \times l_{14});$
$q_3 = (x_2 \times l_{12});$
$q_4 = (x_1 \times l_5);$
$q_5 = (l_3 \times l_7) \oplus x_5;$
$q_6 = (l_4 \times l_{13}) \oplus x_1;$
$q_7 = (l_8 \times l_{10}) \oplus x_1;$
$l_{16} = q_5 \oplus q_6;$
$y_0 = q_1 \oplus q_2 \oplus q_4 \oplus q_6 \oplus l_9;$
$y_1 = q_0 \oplus q_3 \oplus q_7 \oplus x_2;$
$y_2 = q_3 \oplus q_4 \oplus l_{16} \oplus l_1;$
$y_3 = q_0 \oplus q_1 \oplus l_{15};$
$y_4 = q_6 \oplus q_7 \oplus l_2;$
$y_5 = q_0 \oplus l_{16} \oplus l_6;$

X^9
A= 6 X= 28

$l_0 = x_5 \oplus x_4;$
$l_1 = x_5 \oplus x_3;$
$l_2 = x_4 \oplus x_3;$
$l_3 = x_4 \oplus x_2;$
$l_4 = x_3 \oplus x_1;$
$l_5 = x_4 \oplus x_0;$
$l_6 = x_1 \oplus x_0;$
$l_7 = l_0 \oplus x_1;$
$l_8 = l_0 \oplus l_4;$
$l_9 = l_3 \oplus l_8;$
$l_{10} = x_5 \oplus l_9;$
$l_{11} = x_2 \oplus l_5;$
$l_{12} = x_5 \oplus l_3;$
$l_{13} = l_7 \oplus l_{11};$
$q_0 = (x_2 \times l_2);$
$q_1 = (l_0 \times l_4);$
$q_2 = (l_{11} \times l_9) \oplus x_2;$
$y_0 = q_0 \oplus q_1 \oplus q_2 \oplus l_{13};$
$q_3 = (l_1 \times l_3);$
$l_{13} = q_2 \oplus q_3;$
$q_4 = (l_5 \times l_7) \oplus x_4;$
$l_{14} = q_1 \oplus q_4;$
$y_1 = l_{13} \oplus l_{14} \oplus l_{12};$
$q_5 = (x_3 \times l_6);$
$l_{15} = q_0 \oplus q_5;$
$y_5 = l_{14} \oplus l_{15} \oplus l_8;$
$y_3 = 0;$
$y_2 = l_{13} \oplus l_{15} \oplus l_{10};$
$y_4 = y_2;$

BL_ECCZ_1
A= 9 X= 23

$l_0 = x_5 \oplus x_3;$
$l_1 = x_5 \oplus x_2;$
$l_2 = x_3 \oplus x_2;$
$l_3 = x_5 \oplus x_1;$
$l_4 = x_3 \oplus x_0;$
$l_5 = x_4 \oplus l_2;$
$l_6 = x_4 \oplus l_3;$
$l_7 = x_4 \oplus l_4;$
$l_8 = l_0 \oplus l_7;$
$l_9 = x_5 \oplus l_4;$
$l_{10} = x_1 \oplus l_7;$
$l_{11} = l_1 \oplus l_6;$
$q_0 = (x_1 \times l_8);$
$q_1 = (x_0 \times l_3);$
$y_4 = q_0 \oplus q_1;$
$q_2 = (x_4 \times l_0);$
$q_3 = (x_5 \times l_5);$
$y_5 = q_2 \oplus q_3;$
$q_4 = (x_3 \times l_{11});$
$q_5 = (l_6 \times l_9) \oplus x_5;$
$y_3 = q_4 \oplus q_1 \oplus q_5;$
$q_6 = (x_2 \times l_4);$
$q_7 = (l_2 \times l_{10}) \oplus x_3;$
$l_{12} = q_2 \oplus q_6;$
$y_2 = l_{12} \oplus q_7;$
$y_1 = l_{12} \oplus q_4;$
$q_8 = (l_1 \times l_7);$
$y_0 = q_6 \oplus q_8 \oplus q_1;$

BL_ECCZ_8
A= 8 X= 34

$$
\begin{aligned}
l_0 &= x_4 \oplus x_2;\\
l_1 &= x_3 \oplus x_2;\\
l_2 &= x_4 \oplus x_1;\\
l_3 &= x_2 \oplus x_1;\\
l_4 &= x_5 \oplus x_0;\\
l_5 &= x_1 \oplus x_0;\\
l_6 &= x_5 \oplus x_3;\\
l_7 &= l_6 \oplus l_3;\\
l_8 &= l_6 \oplus l_5;\\
l_9 &= l_6 \oplus l_0;\\
l_{10} &= x_5 \oplus l_2;\\
l_{11} &= l_0 \oplus l_4;\\
l_{12} &= l_{10} \oplus l_8;\\
l_{13} &= x_4 \oplus l_8;\\
l_{14} &= x_3 \oplus l_4;\\
q_0 &= (x_0 \times l_2);\\
q_1 &= (l_5 \times l_{10}) \oplus x_1;\\
y_4 &= q_0 \oplus q_1;\\
q_2 &= (x_2 \times l_{14});\\
q_3 &= (l_3 \times l_4);\\
y_3 &= q_2 \oplus q_0 \oplus q_3;\\
q_4 &= (l_1 \times l_{13}) \oplus x_3;\\
q_5 &= (l_9 \times l_8) \oplus l_6;\\
l_{15} &= q_4 \oplus q_5;\\
y_0 &= l_{15} \oplus q_1;\\
q_6 &= (l_0 \times l_{12}) \oplus x_4;\\
y_5 &= q_2 \oplus q_6 \oplus l_{15} \oplus y_4;\\
y_2 &= q_6 \oplus q_5 \oplus y_4;\\
q_7 &= (l_7 \times l_{11}) \oplus x_5 \oplus x_2;\\
y_1 &= q_4 \oplus y_4 \oplus q_3 \oplus q_7;
\end{aligned}
$$

BL_ECCZ_11
A= 8 X= 29

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_5 \oplus x_3;\\
l_2 &= x_4 \oplus x_2;\\
l_3 &= x_3 \oplus x_2;\\
l_4 &= x_3 \oplus x_1;\\
l_5 &= x_3 \oplus x_0;\\
l_6 &= l_0 \oplus l_5;\\
l_7 &= x_5 \oplus l_4;\\
l_8 &= l_2 \oplus x_0;\\
l_9 &= l_3 \oplus l_7;\\
l_{10} &= l_0 \oplus l_8;\\
l_{11} &= x_4 \oplus l_9;\\
l_{12} &= x_5 \oplus x_1;\\
l_{13} &= l_{12} \oplus l_6;\\
q_0 &= (x_3 \times l_0);\\
q_1 &= (x_5 \times l_2);\\
y_5 &= q_0 \oplus q_1;\\
q_2 &= (l_1 \times l_9) \oplus x_5;\\
q_3 &= (x_2 \times l_5);\\
l_{14} &= q_1 \oplus q_3;\\
y_1 &= q_2 \oplus l_{14};\\
q_4 &= (l_3 \times l_{13}) \oplus x_3;\\
l_{15} &= q_0 \oplus q_4;\\
y_2 &= l_{14} \oplus l_{15};\\
q_5 &= (l_4 \times l_{10});\\
y_0 &= q_2 \oplus q_3 \oplus l_{15} \oplus q_5;\\
q_6 &= (l_7 \times l_8);\\
y_4 &= q_0 \oplus q_5 \oplus q_6;\\
q_7 &= (l_{11} \times l_6) \oplus l_0;\\
y_3 &= l_{15} \oplus q_6 \oplus q_7;
\end{aligned}
$$

BL_ECCZ_14
A= 9 X= 23

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_5 \oplus x_3;\\
l_2 &= x_3 \oplus x_2;\\
l_3 &= x_4 \oplus x_1;\\
l_4 &= x_3 \oplus x_0;\\
l_5 &= x_1 \oplus x_0;\\
l_6 &= l_2 \oplus x_1;\\
l_7 &= l_0 \oplus x_2;\\
l_8 &= l_0 \oplus l_5;\\
l_9 &= x_5 \oplus l_4;\\
l_{10} &= x_3 \oplus l_5;\\
l_{11} &= x_2 \oplus l_3;\\
q_0 &= (l_1 \times l_7) \oplus x_5;\\
q_1 &= (x_4 \times l_4);\\
y_3 &= q_0 \oplus q_1;\\
q_2 &= (x_4 \times x_3);\\
q_3 &= (x_5 \times l_{11});\\
y_5 &= q_2 \oplus q_3;\\
q_4 &= (l_0 \times l_{10});\\
y_4 &= q_3 \oplus q_1 \oplus q_4;\\
q_5 &= (x_2 \times l_9);\\
q_6 &= (l_2 \times l_8);\\
l_{12} &= q_0 \oplus q_5;\\
y_2 &= l_{12} \oplus q_6;\\
q_7 &= (x_3 \times l_3);\\
y_1 &= l_{12} \oplus q_7;\\
q_8 &= (l_5 \times l_6) \oplus x_1;\\
y_0 &= q_2 \oplus q_6 \oplus q_8;
\end{aligned}
$$

S_61
A= 6 X= 17

$$
\begin{aligned}
l_0 &= x_5 \oplus x_3;\\
l_1 &= x_2 \oplus x_1;\\
l_2 &= x_3 \oplus l_1;\\
l_3 &= x_3 \oplus x_2;\\
l_4 &= l_3 \oplus x_0;\\
l_5 &= x_5 \oplus l_2;\\
l_6 &= x_4 \oplus l_4;\\
l_7 &= l_5 \oplus l_6;\\
l_8 &= l_0 \oplus l_6;\\
l_9 &= x_0 \oplus l_7;\\
q_0 &= (x_2 \times l_7);\\
y_2 &= q_0;\\
q_1 &= (x_5 \times l_2);\\
y_3 &= q_1;\\
q_2 &= (x_4 \times l_5);\\
q_3 &= (l_0 \times l_8) \oplus x_5;\\
y_0 &= q_2 \oplus q_3;\\
q_4 &= (l_1 \times l_6) \oplus x_2;\\
y_1 &= q_1 \oplus q_0 \oplus q_4;\\
q_5 &= (l_9 \times l_4);\\
y_5 &= q_2 \oplus q_5;\\
y_4 &= q_0 \oplus q_5;
\end{aligned}
$$

S_62
A= 7 X= 24

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_5 \oplus x_2;\\
l_2 &= x_2 \oplus x_0;\\
l_3 &= x_1 \oplus x_0;\\
l_4 &= l_0 \oplus l_3;\\
l_5 &= l_1 \oplus l_3;\\
l_6 &= l_0 \oplus x_1;\\
l_7 &= x_5 \oplus x_3;\\
l_8 &= l_7 \oplus l_4;\\
l_9 &= x_5 \oplus l_5;\\
l_{10} &= l_8 \oplus l_5;\\
l_{11} &= l_7 \oplus l_2;\\
l_{12} &= x_5 \oplus l_{10};\\
q_0 &= (x_2 \times l_4);\\
y_2 &= q_0;\\
q_1 &= (l_0 \times l_{11}) \oplus x_5;\\
q_2 &= (l_{12} \times l_5) \oplus x_2;\\
y_5 &= q_1 \oplus q_2;\\
q_3 &= (l_2 \times l_{10}) \oplus x_2;\\
y_0 &= q_1 \oplus q_3;\\
q_4 &= (x_4 \times l_9);\\
y_1 &= q_4 \oplus q_2 \oplus q_3;\\
q_5 &= (l_3 \times l_6) \oplus x_1;\\
y_4 &= q_4 \oplus q_0 \oplus q_5;\\
q_6 &= (l_1 \times l_8);\\
y_3 &= q_6 \oplus q_5;
\end{aligned}
$$

S-box	LUT
χ_6	0,17,34,19,5,20,38,23,10,27,40,25,13,28,46,31 20,5,54,7,17,0,50,3,26,11,56,9,29,12,62,15,40 41,10,43,45,44,14,47,34,35,0,33,37,36,6,39,52 53,22,55,49,48,18,51,58,59,24,57,61,60,30,63
S_2256	0,1,2,3,4,6,7,5,8,12,16,20,32,39,57,62,9,17,21 13,40,51,53,46,50,47,52,41,63,33,56,38,10,45 27,60,43,15,59,31,58,24,49,19,55,22,61,28,29 35,18,44,25,36,23,42,30,37,11,48,54,14,34,26
S_2257	0,1,2,3,4,6,7,5,8,12,16,20,32,39,57,62,9,17,21 13,41,50,52,47,40,53,46,51,36,58,35,61,10,25 37,54,33,49,15,31,45,59,24,14,42,63,30,11,29 23,44,38,18,27,34,43,19,28,56,55,48,60,26,22
S_2258	0,1,2,3,4,6,7,5,8,12,16,20,32,39,57,62,9,17,21 13,41,50,52,47,55,42,49,44,59,37,60,34,10,25 38,53,35,51,14,30,61,43,11,29,56,45,15,26,22 28,36,46,27,18,40,33,23,24,63,48,54,58,31,19
S_2259	0,1,2,3,4,6,8,10,5,11,16,30,32,45,59,54,7,20,41 58,47,63,15,31,48,44,9,21,57,38,14,17,12,25 24,13,34,52,56,46,40,50,43,49,39,62,42,51,35 36,27,28,33,37,23,19,53,61,26,18,22,29,55,60
S_2260	0,1,2,3,4,6,8,10,5,11,16,30,32,45,59,54,7,24,40 55,48,44,17,13,9,25,49,33,31,12,41,58,14,27 26,15,57,47,35,53,61,39,62,36,43,50,38,63,46 37,23,28,42,34,29,21,22,18,56,60,51,52,19,20
S_2261	0,1,2,3,4,6,8,10,5,11,16,30,32,45,59,54,7,34,21 48,13,43,17,55,56,18,61,23,19,58,24,49,9,52 20,41,31,33,12,50,46,28,36,22,25,40,29,44,62 39,51,42,38,60,37,63,35,53,57,47,26,15,14,27
S_2262	0,1,2,3,4,6,8,10,5,12,16,25,32,42,59,49,7,20,14 29,52,36,51,35,53,46,43,48,39,63,55,47,9,58 44,31,22,38,61,13,19,40,33,26,45,21,17,41,54 23,24,57,30,60,62,28,27,50,34,11,18,56,37,15
S_2263	0,1,2,3,4,8,16,28,5,12,32,41,10,14,57,61,6,62,23 47,33,20,38,19,43,27,29,45,7,58,39,26,9,22,55 40,11,25,35,49,44,59,53,34,37,63,42,48,21,51 56,30,52,31,15,36,24,54,18,60,50,17,46,13
X^3	0,1,8,15,27,14,35,48,53,39,43,63,47,41,1,1,41,15 15,47,52,6,34,22,20,33,36,23,8,41,8,47,36,52,35 53,35,39,20,22,33,34,48,53,39,48,6,23,22,33,63 14,23,52,14,43,27,63,36,6,27,43,20,34
X^9	0,1,53,1,23,1,53,22,52,22,53,34,23,34,1,1,34,1 1,23,23,35,35,34,52,52,35,22,53,34,53,23,35 23,53,52,53,22,52,34,52,35,22,52,22,22,35,22 34,52,34,1,22,23,1,53,23,34,35,35,23,53,52,35
BL_ECCZ_1	0,0,0,1,0,2,4,7,0,4,6,3,8,14,10,13,0,8,16,25,5,15 17,26,32,44,54,59,45,35,63,48,0,16,24,9,32,50 60,47,41,61,55,34,1,23,27,12,15,23,7,30,42,48 38,61,6,26,8,21,43,53,33,62
BL_ECCZ_8	0,0,0,1,0,2,4,7,0,4,6,3,8,14,10,13,0,8,16,25,5,15 17,26,32,44,54,59,45,35,63,48,0,16,26,11,36,54 58,41,33,53,61,40,13,27,21,2,39,63,45,52,6,28 8,19,38,58,42,55,15,17,7,24
BL_ECCZ_11	0,0,0,1,0,2,4,7,0,4,6,3,8,14,10,13,0,8,16,25,5,15 17,26,32,44,54,59,45,35,63,48,0,16,26,11,53,39,43 56,39,51,59,46,26,12,2,21,62,38,52,45,14,20,0,27 57,37,53,40,1,31,9,22
BL_ECCZ_14	0,0,0,1,0,2,4,7,0,4,6,3,8,14,10,13,0,8,16,25,5,15 17,26,32,44,54,59,45,35,63,48,0,16,32,49,61,47 25,10,27,15,61,40,46,56,12,27,62,38,14,23,6 28,50,41,5,25,51,46,53,43,7,24
S_61	0,0,0,50,0,20,36,2,0,1,50,1,3,22,21,50,0,48,35,33,5 33,2,20,16,33,1,2,22,51,35,52,0,49,10,9,45,8 3,20,59,11,3,1,21,49,9,31,32,33,9,58,8,29 5,34,11,11,16,34,32,52,31,57
S_62	0,0,0,24,0,61,62,27,0,33,34,27,41,53,53,49,0,10,40 58,14,57,24,55,33,10,43,24,6,16,50,60,0,50,16 58,20,27,58,45,11,24,57,50,54,24,58,12,11,51 51,19,17,20,23,10,33,56,59, 58,18,54,54,10

7-bit S-box implementations

χ_7

A = 7 X = 14

$$
\begin{aligned}
l_0 &= x_6 \oplus x_4;\\
l_1 &= x_5 \oplus x_3;\\
l_2 &= x_4 \oplus x_2;\\
l_3 &= x_6 \oplus x_1;\\
l_4 &= x_3 \oplus x_1;\\
l_5 &= x_5 \oplus x_0;\\
l_6 &= x_2 \oplus x_0;\\
q_0 &= (x_2 \times x_1);\\
y_0 &= q_0 \oplus l_6;\\
q_1 &= (x_6 \times x_5);\\
y_4 &= q_1 \oplus l_0;\\
q_2 &= (x_5 \times x_4);\\
y_3 &= q_2 \oplus l_1;\\
q_3 &= (x_4 \times x_3);\\
y_2 &= q_3 \oplus l_2;\\
q_4 &= (x_3 \times x_2);\\
y_1 &= q_4 \oplus l_4;\\
q_5 &= (x_6 \times x_0);\\
y_5 &= q_5 \oplus l_5;\\
q_6 &= (x_1 \times x_0);\\
y_6 &= q_6 \oplus l_3;
\end{aligned}
$$

X^3

A = 11 X = 45

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_6 \oplus x_3;\\
l_2 &= x_5 \oplus x_3;\\
l_3 &= x_6 \oplus x_1;\\
l_4 &= x_5 \oplus x_1;\\
l_5 &= x_4 \oplus x_0;\\
l_6 &= l_0 \oplus x_2;\\
l_7 &= x_3 \oplus l_4;\\
l_8 &= l_6 \oplus l_4;\\
l_9 &= x_2 \oplus l_3;\\
l_{10} &= l_1 \oplus x_0;\\
l_{11} &= l_3 \oplus l_8;\\
l_{12} &= x_6 \oplus l_0;\\
l_{13} &= l_2 \oplus l_8;\\
l_{14} &= l_{11} \oplus x_0;\\
l_{15} &= l_6 \oplus x_0;\\
l_{16} &= l_{12} \oplus x_0;\\
l_{17} &= x_6 \oplus l_5;\\
l_{18} &= l_7 \oplus x_0;\\
q_0 &= (x_2 \times l_{12});\\
q_1 &= (x_4 \times l_4);\\
q_2 &= (l_0 \times l_7) \oplus x_5;\\
q_3 &= (l_1 \times l_8);\\
q_4 &= (l_2 \times l_{17});\\
q_5 &= (l_6 \times l_{18}) \oplus x_5;\\
q_6 &= (x_1 \times l_{15});\\
q_7 &= (l_3 \times l_{10}) \oplus x_6;\\
q_8 &= (l_4 \times l_{14});\\
q_9 &= (l_5 \times l_9);\\
q_{10} &= (l_{16} \times l_{13}) \oplus l_0;\\
l_{19} &= q_0 \oplus q_4;\\
l_{20} &= q_2 \oplus q_8;\\
y_0 &= q_1 \oplus q_2 \oplus q_3 \oplus x_0;\\
y_1 &= l_{19} \oplus q_7 \oplus q_9 \oplus q_{10} \oplus x_5;\\
y_2 &= q_0 \oplus l_{20} \oplus q_5 \oplus x_3;\\
y_3 &= q_1 \oplus l_{19} \oplus l_7;\\
y_4 &= q_4 \oplus q_8 \oplus q_{10} \oplus x_6;\\
y_5 &= q_7 \oplus l_{20} \oplus x_4;\\
y_6 &= l_{20} \oplus q_4 \oplus q_6 \oplus l_{11};
\end{aligned}
$$

X^5

A= 11 X= 56

$$
\begin{aligned}
l_0 &= x_6 \oplus x_4;\\
l_1 &= x_6 \oplus x_3;\\
l_2 &= x_5 \oplus x_3;\\
l_3 &= x_6 \oplus x_2;\\
l_4 &= x_5 \oplus x_2;\\
l_5 &= x_3 \oplus x_2;\\
l_6 &= x_5 \oplus x_1;\\
l_7 &= x_4 \oplus x_1;\\
l_8 &= x_5 \oplus x_0;\\
l_9 &= x_2 \oplus l_6;\\
l_{10} &= x_4 \oplus l_4;\\
l_{11} &= x_6 \oplus l_7;\\
l_{12} &= l_2 \oplus l_9;\\
l_{13} &= x_5 \oplus l_0;\\
l_{14} &= x_5 \oplus l_{11};\\
l_{15} &= l_0 \oplus x_0;\\
l_{16} &= x_4 \oplus l_2;\\
l_{17} &= x_3 \oplus l_8;\\
l_{18} &= l_1 \oplus x_0;\\
l_{19} &= x_5 \oplus l_5;\\
l_{20} &= l_7 \oplus x_0;\\
l_{21} &= x_2 \oplus l_{15};\\
l_{22} &= x_4 \oplus l_{17};\\
l_{23} &= l_3 \oplus l_{18};\\
l_{24} &= x_6 \oplus l_{20};\\
q_0 &= (x_3 \times l_9);\\
q_1 &= (l_3 \times l_{11}) \oplus x_6;\\
q_2 &= (l_{19} \times l_{14}) \oplus x_5;\\
y_0 &= q_0 \oplus q_1 \oplus q_2 \oplus l_8;\\
q_3 &= (l_6 \times l_{21});\\
q_4 &= (l_7 \times l_{18});\\
l_{25} &= q_3 \oplus q_4;\\
y_2 &= q_0 \oplus l_{25} \oplus l_0;\\
q_5 &= (x_1 \times l_1);\\
q_6 &= (l_{15} \times l_{17}) \oplus x_0;\\
l_{26} &= q_5 \oplus q_6;\\
y_3 &= l_{26} \oplus l_{25} \oplus l_5;\\
q_7 &= (l_{10} \times l_{24}) \oplus x_4;\\
l_{27} &= q_1 \oplus q_7;\\
q_8 &= (l_{12} \times l_{22}) \oplus x_3;\\
y_6 &= q_0 \oplus q_7 \oplus q_4 \oplus q_8 \oplus l_{13};\\
q_9 &= (x_4 \times l_2);\\
l_{28} &= q_8 \oplus q_9;\\
y_4 &= l_{27} \oplus l_{26} \oplus l_{28} \oplus l_4;\\
q_{10} &= (l_{23} \times l_{20}) \oplus x_0;\\
y_5 &= l_{28} \oplus l_{26} \oplus q_2 \oplus q_{10} \oplus l_6;\\
y_1 &= q_5 \oplus l_{27} \oplus q_4 \oplus l_{16};
\end{aligned}
$$

X^9

A = 11 X = 53

$$
\begin{aligned}
l_0 &= x_6 \oplus x_4;\\
l_1 &= x_5 \oplus x_4;\\
l_2 &= x_6 \oplus x_3;\\
l_3 &= x_3 \oplus x_2;\\
l_4 &= x_5 \oplus x_1;\\
l_5 &= x_3 \oplus x_1;\\
l_6 &= x_2 \oplus x_1;\\
l_7 &= x_3 \oplus x_0;\\
l_8 &= x_5 \oplus l_5;\\
l_9 &= x_5 \oplus x_2;\\
l_{10} &= l_9 \oplus x_0;\\
l_{11} &= l_1 \oplus x_0;\\
l_{12} &= x_4 \oplus l_3;\\
l_{13} &= l_{12} \oplus x_1;\\
l_{14} &= x_4 \oplus l_4;\\
l_{15} &= x_6 \oplus l_1;\\
l_{16} &= x_6 \oplus l_6;\\
l_{17} &= x_2 \oplus l_7;\\
l_{18} &= x_6 \oplus l_5;\\
l_{19} &= l_4 \oplus l_{11};\\
l_{20} &= x_6 \oplus x_5;\\
l_{21} &= l_{20} \oplus l_7;\\
l_{22} &= l_{20} \oplus l_{13};\\
l_{23} &= l_{20} \oplus l_{12};\\
q_0 &= (x_5 \times l_{13});\\
q_1 &= (x_2 \times l_{18});\\
q_2 &= (l_8 \times l_{16}) \oplus x_1;\\
l_{24} &= q_1 \oplus q_2;\\
y_0 &= q_0 \oplus l_{24} \oplus l_{21};\\
q_3 &= (l_0 \times l_6);\\
q_4 &= (x_6 \times l_{10});\\
l_{25} &= q_3 \oplus q_4;\\
y_3 &= q_0 \oplus l_{25} \oplus l_5;\\
q_5 &= (l_3 \times l_4);\\
q_6 &= (x_4 \times l_{17});\\
l_{26} &= q_1 \oplus q_5;\\
l_{27} &= l_{26} \oplus q_6;\\
y_1 &= q_4 \oplus l_{27} \oplus l_1;\\
q_7 &= (l_2 \times l_{11});\\
y_5 &= l_{24} \oplus q_5 \oplus q_7 \oplus l_{15};\\
q_8 &= (x_0 \times l_{15});\\
y_6 &= l_{25} \oplus l_{27} \oplus q_8 \oplus q_7 \oplus l_{12};\\
q_9 &= (l_{19} \times l_{23}) \oplus x_4;\\
y_4 &= q_3 \oplus l_{24} \oplus q_6 \oplus q_9 \oplus x_2;\\
q_{10} &= (l_7 \times l_{22}) \oplus x_3;\\
y_2 &= l_{24} \oplus q_5 \oplus q_8 \oplus q_9 \oplus q_{10} \oplus l_{14};
\end{aligned}
$$

X^{17}

A= 11 X= 54

$$
\begin{aligned}
l_0 &= x_6 \oplus x_4;\\
l_1 &= x_5 \oplus x_3;\\
l_2 &= x_3 \oplus x_2;\\
l_3 &= x_6 \oplus x_1;\\
l_4 &= x_3 \oplus x_1;\\
l_5 &= l_2 \oplus l_3;\\
l_6 &= l_5 \oplus x_0;\\
l_7 &= x_5 \oplus l_2;\\
l_8 &= l_7 \oplus l_6;\\
l_9 &= l_4 \oplus l_8;\\
l_{10} &= l_7 \oplus x_1;\\
l_{11} &= l_0 \oplus l_1;\\
l_{12} &= x_4 \oplus l_2;\\
l_{13} &= l_{12} \oplus l_9;\\
l_{14} &= l_1 \oplus l_{13};\\
l_{15} &= l_{11} \oplus l_{12};\\
l_{16} &= l_0 \oplus l_{15};\\
l_{17} &= l_{11} \oplus x_1;\\
l_{18} &= l_1 \oplus l_5;\\
l_{19} &= l_7 \oplus l_9;\\
l_{20} &= x_6 \oplus l_7;\\
l_{21} &= l_{17} \oplus l_{14};\\
l_{22} &= l_{11} \oplus l_{10};\\
l_{23} &= l_{15} \oplus l_{22};\\
l_{24} &= l_{20} \oplus l_{14};\\
q_0 &= (l_{15} \times l_{12}) \oplus x_2;\\
q_1 &= (l_4 \times l_{16});\\
q_2 &= (l_{23} \times l_5) \oplus x_1;\\
l_{25} &= q_0 \oplus q_2;\\
y_0 &= q_1 \oplus l_{25} \oplus l_9;\\
q_3 &= (l_1 \times l_3);\\
q_4 &= (x_5 \times l_6);\\
l_{26} &= q_1 \oplus q_3;\\
y_6 &= l_{26} \oplus q_4 \oplus l_{20};\\
q_5 &= (l_7 \times l_{22}) \oplus x_2;\\
q_6 &= (x_6 \times l_{21});\\
l_{27} &= q_4 \oplus q_5;\\
y_3 &= l_{27} \oplus q_6 \oplus l_{18};\\
q_7 &= (x_3 \times l_8);\\
y_5 &= q_0 \oplus l_{26} \oplus q_7 \oplus l_{17};\\
q_8 &= (x_1 \times l_{13});\\
l_{28} &= q_7 \oplus q_8;\\
y_2 &= l_{25} \oplus l_{27} \oplus l_{28} \oplus l_{11};\\
q_9 &= (l_{10} \times l_{14}) \oplus l_2;\\
y_4 &= l_{25} \oplus l_{27} \oplus q_9 \oplus l_2;\\
q_{10} &= (l_{24} \times l_{19}) \oplus x_0;\\
y_1 &= q_1 \oplus q_5 \oplus q_2 \oplus l_{28} \oplus q_{10} \oplus l_0;
\end{aligned}
$$

X^{33}

A= 11 X= 58

$$
\begin{aligned}
l_0 &= x_6 \oplus x_4;\\
l_1 &= x_5 \oplus x_4;\\
l_2 &= x_5 \oplus x_3;\\
l_3 &= x_5 \oplus x_2;\\
l_4 &= x_3 \oplus x_2;\\
l_5 &= x_4 \oplus x_1;\\
l_6 &= x_2 \oplus x_1;\\
l_7 &= x_5 \oplus x_0;\\
l_8 &= x_1 \oplus x_0;\\
l_9 &= x_6 \oplus l_6;\\
l_{10} &= x_5 \oplus l_4;\\
l_{11} &= l_2 \oplus l_6;\\
l_{12} &= l_9 \oplus l_8;\\
l_{13} &= x_4 \oplus l_8;\\
l_{14} &= l_2 \oplus l_5;\\
l_{15} &= l_0 \oplus l_{14};\\
l_{16} &= x_6 \oplus l_2;\\
l_{17} &= x_6 \oplus l_{10};\\
l_{18} &= l_2 \oplus l_8;\\
l_{19} &= x_6 \oplus l_5;\\
l_{20} &= x_5 \oplus l_{19};\\
l_{21} &= x_5 \oplus l_{11};\\
l_{22} &= x_5 \oplus l_{15};\\
l_{23} &= x_5 \oplus l_9;\\
l_{24} &= l_1 \oplus l_{18};\\
q_0 &= (x_1 \times l_1);\\
q_1 &= (l_5 \times l_{10});\\
q_2 &= (l_9 \times l_{14}) \oplus x_1;\\
l_{25} &= q_0 \oplus q_2;\\
y_0 &= q_1 \oplus l_{25} \oplus l_7;\\
q_3 &= (l_0 \times l_{22}) \oplus x_6;\\
q_4 &= (x_6 \times x_0);\\
l_{26} &= q_3 \oplus q_4;\\
y_5 &= l_{25} \oplus l_{26} \oplus l_{19};\\
q_5 &= (l_2 \times l_6);\\
q_6 &= (l_{13} \times l_{23}) \oplus x_1;\\
y_2 &= q_0 \oplus q_5 \oplus q_6 \oplus l_{15};\\
q_7 &= (l_3 \times l_{18}) \oplus x_5;\\
q_8 &= (l_8 \times l_{20}) \oplus x_1;\\
l_{27} &= q_1 \oplus q_7;\\
y_4 &= q_0 \oplus q_4 \oplus q_7 \oplus q_8 \oplus l_{16};\\
q_9 &= (l_{17} \times l_{24}) \oplus x_3;\\
l_{28} &= q_6 \oplus q_9;\\
y_6 &= q_0 \oplus l_{27} \oplus q_9 \oplus l_4;\\
q_{10} &= (l_{21} \times l_{12}) \oplus x_2;\\
y_3 &= q_3 \oplus l_{27} \oplus l_{28} \oplus q_{10} \oplus l_{11};\\
y_1 &= l_{26} \oplus l_{27} \oplus q_8 \oplus l_{28} \oplus x_6;
\end{aligned}
$$

X^{65}

A = 11 X = 52

$$
\begin{aligned}
l_0 &= x_5 \oplus x_3;\\
l_1 &= x_6 \oplus x_2;\\
l_2 &= x_3 \oplus x_2;\\
l_3 &= x_3 \oplus x_1;\\
l_4 &= x_6 \oplus x_0;\\
l_5 &= x_4 \oplus x_0;\\
l_6 &= x_3 \oplus l_5;\\
l_7 &= l_1 \oplus l_5;\\
l_8 &= l_3 \oplus x_0;\\
l_9 &= x_5 \oplus l_3;\\
l_{10} &= l_1 \oplus l_3;\\
l_{11} &= x_5 \oplus l_1;\\
l_{12} &= x_6 \oplus l_3;\\
l_{13} &= x_6 \oplus l_0;\\
l_{14} &= l_2 \oplus l_8;\\
l_{15} &= l_5 \oplus l_{14};\\
l_{16} &= x_6 \oplus l_9;\\
l_{17} &= x_6 \oplus l_{15};\\
l_{18} &= x_4 \oplus l_9;\\
l_{19} &= l_{15} \oplus l_{10};\\
l_{20} &= l_0 \oplus l_{19};\\
q_0 &= (x_4 \times l_0);\\
q_1 &= (l_{11} \times l_{15}) \oplus x_2;\\
q_2 &= (l_2 \times l_{18}) \oplus x_3;\\
l_{21} &= q_1 \oplus q_2;\\
y_0 &= q_0 \oplus l_{21} \oplus x_0;\\
q_3 &= (l_3 \times l_5);\\
q_4 &= (l_{16} \times l_7) \oplus x_6;\\
y_6 &= l_{21} \oplus q_3 \oplus q_4 \oplus x_4;\\
q_5 &= (l_{20} \times l_{10}) \oplus x_6;\\
q_6 &= (l_{13} \times l_8) \oplus x_3;\\
l_{22} &= q_5 \oplus q_6;\\
y_3 &= q_1 \oplus l_{22} \oplus l_1;\\
q_7 &= (l_{12} \times l_6) \oplus x_3;\\
y_5 &= l_{22} \oplus q_3 \oplus q_7 \oplus l_9;\\
q_8 &= (l_{19} \times l_{14});\\
q_9 &= (l_4 \times l_{17}) \oplus x_6;\\
y_1 &= q_5 \oplus q_8 \oplus q_3 \oplus q_9 \oplus l_0;\\
q_{10} &= (x_2 \times l_9);\\
y_2 &= q_0 \oplus q_{10} \oplus l_{21} \oplus q_7 \oplus q_9 \oplus l_{16};\\
y_4 &= l_{21} \oplus q_8 \oplus q_7 \oplus x_5;
\end{aligned}
$$

$X^3 \oplus X^5$

A = 10 X = 57

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_4 \oplus x_3;\\
l_2 &= x_5 \oplus x_2;\\
l_3 &= x_5 \oplus x_1;\\
l_4 &= x_5 \oplus x_0;\\
l_5 &= x_4 \oplus x_0;\\
l_6 &= x_3 \oplus x_0;\\
l_7 &= x_6 \oplus l_2;\\
l_8 &= l_7 \oplus l_4;\\
l_9 &= x_2 \oplus l_5;\\
l_{10} &= x_2 \oplus l_3;\\
l_{11} &= x_5 \oplus l_1;\\
l_{12} &= x_6 \oplus l_5;\\
l_{13} &= l_1 \oplus l_3;\\
l_{14} &= l_{13} \oplus l_4;\\
l_{15} &= l_7 \oplus l_3;\\
l_{16} &= x_6 \oplus l_0;\\
l_{17} &= l_0 \oplus l_{10};\\
l_{18} &= x_6 \oplus l_3;\\
l_{19} &= x_4 \oplus l_3;\\
l_{20} &= l_{10} \oplus l_{14};\\
l_{21} &= x_4 \oplus l_7;\\
l_{22} &= x_6 \oplus l_1;\\
l_{23} &= x_5 \oplus l_{21};\\
l_{24} &= l_1 \oplus l_{17};\\
q_0 &= (l_0 \times l_6);\\
q_1 &= (l_{17} \times l_8) \oplus x_2;\\
l_{25} &= q_0 \oplus q_1;\\
y_2 &= l_{25} \oplus l_{22};\\
q_2 &= (x_5 \times l_{24});\\
q_3 &= (l_4 \times l_{15});\\
y_3 &= q_2 \oplus q_3 \oplus l_{25} \oplus l_{10};\\
q_4 &= (l_5 \times l_{16}) \oplus x_4;\\
q_5 &= (l_{18} \times l_{20}) \oplus x_5;\\
l_{26} &= q_2 \oplus q_4;\\
y_1 &= l_{25} \oplus l_{26} \oplus q_5 \oplus l_1;\\
q_6 &= (l_3 \times l_9);\\
y_0 &= q_0 \oplus l_{26} \oplus q_6 \oplus q_5 \oplus x_5;\\
q_7 &= (l_{21} \times l_{14}) \oplus x_4;\\
l_{27} &= q_3 \oplus q_7;\\
y_4 &= l_{26} \oplus l_{27} \oplus q_6 \oplus q_1 \oplus l_7;\\
q_8 &= (l_{23} \times l_{13}) \oplus x_4;\\
l_{28} &= q_5 \oplus q_8;\\
y_6 &= l_{26} \oplus l_{28} \oplus l_{27} \oplus l_2;\\
q_9 &= (l_{11} \times l_{12}) \oplus x_4;\\
y_5 &= l_{28} \oplus q_9 \oplus q_6 \oplus q_4 \oplus l_{19};
\end{aligned}
$$

S_71

A = 10 X = 51

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_5 \oplus x_2;\\
l_2 &= x_4 \oplus x_1;\\
l_3 &= l_2 \oplus x_0;\\
l_4 &= x_6 \oplus l_3;\\
l_5 &= x_4 \oplus l_1;\\
l_6 &= l_5 \oplus l_3;\\
l_7 &= x_5 \oplus l_6;\\
l_8 &= x_3 \oplus l_5;\\
l_9 &= x_3 \oplus l_4;\\
l_{10} &= l_2 \oplus l_9;\\
l_{11} &= x_5 \oplus l_4;\\
l_{12} &= x_2 \oplus l_{10};\\
l_{13} &= l_2 \oplus l_{11};\\
l_{14} &= x_4 \oplus l_{13};\\
l_{15} &= x_5 \oplus l_{14};\\
l_{16} &= x_0 \oplus l_9;\\
l_{17} &= x_6 \oplus l_1;\\
l_{18} &= x_0 \oplus l_{12};\\
q_0 &= (l_{15} \times l_8) \oplus x_4;\\
q_1 &= (l_2 \times l_{12});\\
l_{19} &= q_0 \oplus q_1;\\
y_0 &= l_{19} \oplus x_0;\\
q_2 &= (l_1 \times l_4);\\
q_3 &= (l_{16} \times l_{14}) \oplus x_6 \oplus x_4;\\
q_4 &= (l_{10} \times l_6) \oplus x_0;\\
l_{20} &= q_2 \oplus q_4;\\
y_2 &= q_3 \oplus l_{20} \oplus x_2;\\
q_5 &= (l_0 \times l_{18});\\
q_6 &= (l_{17} \times l_7) \oplus x_2;\\
l_{21} &= q_1 \oplus q_5;\\
l_{22} &= q_2 \oplus q_6;\\
y_5 &= l_{21} \oplus q_6 \oplus q_4 \oplus x_5;\\
y_1 &= l_{20} \oplus l_{19} \oplus q_5 \oplus x_1;\\
q_7 &= (l_{18} \times l_{11}) \oplus x_6;\\
y_4 &= l_{22} \oplus q_7 \oplus l_{19} \oplus x_4;\\
q_8 &= (x_5 \times l_9);\\
l_{23} &= q_7 \oplus q_8;\\
y_6 &= l_{21} \oplus l_{22} \oplus l_{23} \oplus q_3 \oplus x_6;\\
q_9 &= (l_{13} \times l_3) \oplus x_0;\\
y_3 &= q_2 \oplus l_{23} \oplus q_0 \oplus q_9 \oplus x_3;
\end{aligned}
$$

S_72
A = 10 X = 48

$$
\begin{aligned}
l_0 &= x_3 \oplus x_2;\\
l_1 &= x_4 \oplus x_1;\\
l_2 &= x_3 \oplus x_1;\\
l_3 &= x_6 \oplus x_0;\\
l_4 &= x_4 \oplus x_0;\\
l_5 &= x_3 \oplus x_0;\\
l_6 &= x_1 \oplus x_0;\\
l_7 &= x_5 \oplus x_2;\\
l_8 &= l_7 \oplus l_1;\\
l_9 &= x_6 \oplus l_0;\\
l_{10} &= x_4 \oplus l_9;\\
l_{11} &= x_6 \oplus l_{10};\\
l_{12} &= x_5 \oplus l_0;\\
l_{13} &= x_5 \oplus l_{11};\\
l_{14} &= x_2 \oplus l_{10};\\
l_{15} &= x_6 \oplus l_4;\\
l_{16} &= l_7 \oplus l_9;\\
l_{17} &= x_2 \oplus l_6;\\
l_{18} &= l_7 \oplus l_2;\\
q_0 &= (x_5 \times l_2);\\
q_1 &= (l_5 \times l_{12}) \oplus x_3;\\
q_2 &= (x_0 \times l_{13});\\
l_{19} &= q_1 \oplus q_2;\\
y_0 &= q_0 \oplus l_{19} \oplus x_0;\\
q_3 &= (l_0 \times l_{14}) \oplus x_3;\\
q_4 &= (l_{11} \times l_{15}) \oplus x_4;\\
l_{20} &= q_0 \oplus q_4;\\
y_5 &= q_3 \oplus q_4 \oplus q_2 \oplus x_5;\\
q_5 &= (l_1 \times l_{16});\\
q_6 &= (l_9 \times l_{17}) \oplus x_2;\\
y_4 &= l_{20} \oplus q_6 \oplus q_5 \oplus l_{19} \oplus x_4;\\
q_7 &= (l_6 \times l_{10});\\
l_{21} &= q_6 \oplus q_7;\\
y_2 &= l_{19} \oplus l_{21} \oplus x_2;\\
y_3 &= q_3 \oplus l_{20} \oplus l_{19} \oplus x_3;\\
q_8 &= (l_4 \times l_{18});\\
y_1 &= q_3 \oplus l_{21} \oplus q_2 \oplus q_8 \oplus x_1;\\
q_9 &= (l_3 \times l_8);\\
y_6 &= q_0 \oplus q_5 \oplus l_{21} \oplus q_1 \oplus q_9 \oplus q_8 \oplus x_6;
\end{aligned}
$$

S_73
A = 11 X = 50

$$
\begin{aligned}
l_0 &= x_6 \oplus x_4;\\
l_1 &= x_6 \oplus x_3;\\
l_2 &= x_6 \oplus x_2;\\
l_3 &= x_5 \oplus x_0;\\
l_4 &= x_3 \oplus x_0;\\
l_5 &= x_1 \oplus x_0;\\
l_6 &= l_1 \oplus l_3;\\
l_7 &= l_0 \oplus x_1;\\
l_8 &= x_2 \oplus l_7;\\
l_9 &= x_5 \oplus l_1;\\
l_{10} &= l_9 \oplus l_8;\\
l_{11} &= x_4 \oplus l_9;\\
l_{12} &= x_5 \oplus l_8;\\
l_{13} &= l_0 \oplus l_{12};\\
l_{14} &= x_1 \oplus l_{12};\\
l_{15} &= x_3 \oplus l_7;\\
l_{16} &= l_9 \oplus l_2;\\
l_{17} &= l_0 \oplus x_0;\\
l_{18} &= l_{11} \oplus l_{12};\\
l_{19} &= x_6 \oplus l_{14};\\
l_{20} &= x_5 \oplus l_5;\\
l_{21} &= x_6 \oplus l_{11};\\
l_{22} &= x_2 \oplus l_6;\\
q_0 &= (l_9 \times l_{19}) \oplus x_5;\\
q_1 &= (x_0 \times l_{12});\\
y_2 &= q_0 \oplus q_1 \oplus l_{10} \oplus 1;\\
q_2 &= (l_1 \times l_8) \oplus x_6;\\
q_3 &= (l_7 \times l_{13}) \oplus x_1;\\
l_{23} &= q_0 \oplus q_2;\\
y_5 &= l_{23} \oplus q_3 \oplus l_{16} \oplus 1;\\
q_4 &= (x_4 \times l_2);\\
q_5 &= (l_0 \times l_6) \oplus x_6;\\
y_0 &= q_4 \oplus q_0 \oplus q_5 \oplus l_1 \oplus 1;\\
q_6 &= (l_2 \times l_{21});\\
q_7 &= (l_{20} \times l_{11}) \oplus x_5;\\
l_{25} &= q_5 \oplus q_7;\\
y_6 &= q_6 \oplus q_2 \oplus l_{25} \oplus l_{14} \oplus 1;\\
q_8 &= (x_1 \times l_{22});\\
q_9 &= (l_3 \times l_{15});\\
l_{24} &= q_3 \oplus q_8;\\
y_3 &= q_2 \oplus l_{24} \oplus q_9;\\
q_{10} &= (l_4 \times l_{18}) \oplus x_3;\\
y_4 &= l_{23} \oplus l_{25} \oplus q_8 \oplus q_{10} \oplus l_5 \oplus 1;\\
y_1 &= q_4 \oplus l_{23} \oplus l_{24} \oplus l_{17};
\end{aligned}
$$

S_74
A = 7 X = 30

$$
\begin{aligned}
l_0 &= x_5 \oplus x_0;\\
l_1 &= x_5 \oplus x_4;\\
l_2 &= l_1 \oplus x_1;\\
l_3 &= x_4 \oplus x_2;\\
l_4 &= l_3 \oplus x_0;\\
l_5 &= x_6 \oplus l_4;\\
l_6 &= l_3 \oplus l_2;\\
l_7 &= x_3 \oplus l_6;\\
l_8 &= x_6 \oplus x_4;\\
l_9 &= l_8 \oplus x_3;\\
l_{10} &= l_1 \oplus l_7;\\
l_{11} &= l_{10} \oplus l_4;\\
l_{12} &= x_3 \oplus l_4;\\
l_{13} &= x_6 \oplus l_6;\\
l_{14} &= l_1 \oplus l_9;\\
q_0 &= (x_6 \times l_6);\\
q_1 &= (l_7 \times l_{12}) \oplus x_3 \oplus x_2;\\
y_1 &= q_0 \oplus q_1;\\
q_2 &= (l_{14} \times l_2) \oplus x_5;\\
q_3 &= (l_0 \times l_9);\\
y_3 &= q_2 \oplus q_3 \oplus q_1;\\
q_4 &= (l_4 \times l_{11}) \oplus x_0;\\
l_{15} &= q_0 \oplus q_4;\\
y_0 &= q_2 \oplus l_{15};\\
q_5 &= (x_3 \times l_5);\\
y_5 &= l_{15} \oplus q_5;\\
q_6 &= (x_0 \times l_{13});\\
l_{16} &= q_5 \oplus q_6;\\
y_6 &= l_{16} \oplus q_1;\\
y_2 &= l_{15} \oplus q_6;\\
y_4 &= q_0 \oplus q_3 \oplus l_{16};
\end{aligned}
$$

S_75
A= 8 X= 31

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_6 \oplus x_2;\\
l_2 &= x_3 \oplus x_1;\\
l_3 &= x_5 \oplus x_0;\\
l_4 &= x_1 \oplus x_0;\\
l_5 &= x_4 \oplus x_2;\\
l_6 &= l_5 \oplus l_3;\\
l_7 &= x_6 \oplus l_0;\\
l_8 &= l_7 \oplus x_3;\\
l_9 &= x_5 \oplus l_6;\\
l_{10} &= x_6 \oplus l_9;\\
l_{11} &= l_0 \oplus x_1;\\
l_{12} &= l_5 \oplus l_{11};\\
l_{13} &= x_3 \oplus l_4;\\
l_{14} &= x_6 \oplus x_5;\\
l_{15} &= l_{14} \oplus l_{11};\\
l_{16} &= l_2 \oplus l_{12};\\
q_0 &= (x_3 \times l_{10});\\
q_1 &= (l_3 \times l_{15});\\
y_1 &= q_0 \oplus q_1;\\
q_2 &= (l_{12} \times l_8) \oplus x_5;\\
q_3 &= (l_{16} \times l_{11}) \oplus x_5;\\
l_{17} &= q_0 \oplus q_2;\\
y_4 &= l_{17} \oplus q_3;\\
q_4 &= (l_0 \times l_4);\\
y_6 &= l_{17} \oplus q_4;\\
q_5 &= (l_9 \times l_{13}) \oplus x_0;\\
l_{18} &= q_1 \oplus q_5;\\
y_3 &= l_{18};\\
q_6 &= (l_2 \times l_6);\\
y_5 &= q_3 \oplus q_4 \oplus l_{18} \oplus q_6;\\
q_7 &= (l_1 \times l_{11});\\
l_{19} &= q_5 \oplus q_7;\\
y_2 &= q_4 \oplus l_{19};\\
y_0 &= q_0 \oplus q_3 \oplus l_{19};
\end{aligned}
$$

S_76
A= 10 X= 43

$$
\begin{aligned}
l_0 &= x_6 \oplus x_5;\\
l_1 &= x_5 \oplus x_4;\\
l_2 &= x_6 \oplus x_2;\\
l_3 &= x_3 \oplus x_2;\\
l_4 &= x_5 \oplus x_0;\\
l_5 &= l_1 \oplus x_3;\\
l_6 &= l_5 \oplus x_1;\\
l_7 &= x_6 \oplus l_1;\\
l_8 &= x_4 \oplus x_2;\\
l_9 &= l_8 \oplus x_0;\\
l_{10} &= l_0 \oplus x_1;\\
l_{11} &= l_7 \oplus l_6;\\
l_{12} &= l_5 \oplus x_0;\\
l_{13} &= l_3 \oplus l_6;\\
l_{14} &= l_{10} \oplus l_4;\\
l_{15} &= x_2 \oplus l_{11};\\
l_{16} &= x_6 \oplus l_8;\\
l_{17} &= l_0 \oplus l_5;\\
l_{18} &= x_3 \oplus l_{14};\\
l_{19} &= x_6 \oplus l_3;\\
l_{20} &= l_2 \oplus l_{14};\\
q_0 &= (l_{16} \times l_6) \oplus x_4;\\
q_1 &= (l_2 \times l_4);\\
y_2 &= q_0 \oplus q_1;\\
q_2 &= (l_1 \times l_{18});\\
q_3 &= (l_{17} \times l_{14}) \oplus x_6;\\
y_3 &= q_2 \oplus q_3;\\
q_4 &= (l_7 \times l_{19}) \oplus x_6;\\
l_{21} &= q_2 \oplus q_4;\\
q_5 &= (l_{13} \times l_{12}) \oplus l_1;\\
y_0 &= l_{21} \oplus q_1 \oplus q_5;\\
q_6 &= (l_0 \times l_{15}) \oplus x_6;\\
q_7 &= (l_{11} \times l_{20}) \oplus x_1;\\
l_{22} &= q_6 \oplus q_7;\\
y_1 &= l_{22} \oplus y_3;\\
q_8 &= (l_3 \times l_5) \oplus x_3;\\
y_4 &= l_{21} \oplus q_8 \oplus q_7;\\
q_9 &= (l_{10} \times l_9);\\
y_6 &= q_6 \oplus y_2 \oplus q_9;\\
y_5 &= l_{22} \oplus y_3 \oplus q_9 \oplus q_4 \oplus q_5;
\end{aligned}
$$

S-box	LUT
χ_7	0,33,66,35,5,36,70,39,10,43,72,41,13,44,78,47,20,53,86,55,17,48,82,51,26,59 88,57,29,60,94,63,40,9,106,11,45,12,110,15,34,3,96,1,37,4,102,7,52,21,118,23,49,16 114,19,58,27,120,25,61,28,126,31,80,81,18,83,85,84,22,87,90,91,24,89,93,92,30,95,68 69,6,71,65,64,2,67,74,75,8,73,77,76,14,79,104,105,42,107,109,108,46,111,98,99,32,97 101,100,38,103,116,117,54,119,113,112,50,115,122,123,56,121,125,124,62,127
X^3	0,1,8,15,64,85,120,107,12,69,39,104,73,20,82,9,96,119,36,53,62,61,74,79,68,27,35,122 31,84,72,5,10,51,49,14,38,11,45,6,117,4,109,26,92,57,116,23,44,3,91,114,30,37,89,100 123,28,47,78,76,63, 40,93,80,113,29,58,13,56,112,67,54,95,88,55,110,19,48,75,33,22 32,17,98,65,83,118,111,16,77,52,41,66,59,86,102,127,24,7,87,90,25,18,115,34,46,121 71,2,42,105,81,94,99,106,126,101,124,97,108,43,125,60,70,21,103,50
X^5	0,1,32,51,24,27,116,101,10,99,106,17,39,76,11,114,70,67,72,95,110,105,44,57,123,22,53 74,102,9,100,25,115,20,80,37,54,83,89,46,109,98,14,19,29,16,50,45,66,33,79,62,55,86 118,5,107,96,38,63,43,34,42,49,68,125,15,36,60,7,59,18,87,6,92,31,26,73,93,28,13,48 104,71,69,122,108,65,41,124,12,75,84,3,61,120,112,47,56,117,85,8,81,30,119,64,127,90 103,82,35,4,78,21,40,97,91,2,113,58,126,77,88,121,94,111,52,23
X^9	0,1,12,9,80,65,108,121,73,56,6,115,95,62,32,69,102,97,38,37,40,63,88,75,13,122,14,125 5,98,54,85,39,114,3,82,45,104,57,120,101,64,2,35,41,28,126,79,92,15,52,99,72,11,16,87 60,31,23,48,110,93,117,66,33,94,84,47,20,123,81,58,89,86,111,100,42,53,44,55,51,74,10 119,24,113,17,124,105,96,19,30,4,29,78,83,25,50,68,107,118,77,27,36,106,49,116,43,67 8,109,34,22,59,7,46,103,90,70,127,71,26,21,76,112,61,18,91
X^{17}	0,1,40,63,88,75,12,9,117,66,2,35,101,64,110,93,38,37,108,121,80,65,102,97,41,28,60 31,23,48,126,79,109,34,21,76,71,26,67,8,51,74,20,123,81,58,10,119,112,61,106,49,116 43,18,91,84,47,17,124,24,113,33,94,111,100,78,83,4,29,89,86,103,90,25,50,68,107,70 127,105,96,42,53,44,55,19,30,27,36,7,46,22,59,118,77,45,104,92,15,52,99,57,120,14,125 32,69,95,62,13,122,16,87,3,82,39,114,72,11,73,56,5,98,54,85,6,115
X^{33}	0,1,44,57,72,95,24,27,92,31,84,3,41,124,93,28,32,51,110,105,70,67,116,101,87,6,61 120,12,75,26,73,29,16,38,63,42,49,109,98,81,30,78,21,91,2,56,117,50,45,107,96,43,34 14,19,85,8,40,97,113,58,112,47,54,83,79,62,118,5,115,20,35,4,126,77,94,111,127,90,89 46,66,33,55,86,80,37,103,82,88,121,52,23,119,64,10,99,100,25,53,74,39,76,15,36,69,122 13,48,59,18,106,17,102,9,123,22,11,114,68,125,108,65,104,71,60,7
X^{65}	0,1,36,53,8,15,96,119,38,11,89,100,45,6,30,37,64,85,74,79,120,107,62,61,10,51,91 114,49,14,44,3,54,95,77,52,88,55,111,16,71,2,103,50,42,105,70,21,110,19,59,86,48,75 41,66,115,34,125,60,46,121,108,43,12,69,35,122,39,104,68,27,92,57,40,93,116,23,76,63 73,20,72,5,82,9,31,84,117,4,47,78,109,26,123,28,80,113,32,17,29,58,33,22,87,90,124,97 25,18,126,101,13,56,83,118,112,67,98,65,102,127,99,106,24,7,81,94
$X^3 \oplus X^5$	0,0,40,60,88,78,12,14,6,38,77,121,110,88,89,123,38,52,108,106,80,84,102 118,63,13,22,48,121,93,44,28,121,39,97,43,16,88,116,40,24,102,99,9,65,41,70,58,110,34 20,76,41,115,47,97,16,124,9,113,103,29,2,108,20,12,18,30,49,63,75,81,97,89,4,40,116,90 109,87,44,38,72,86,39,59,63,55,70,108,65,127,125,65,6,46,22,80,32,114,2,82,72,12,4,98 81,35,32,80,9,109,31,75,75,11,37,103,13,91,18,102,37,69,24,122,83,37
S_71	0,1,2,26,4,78,3,80,8,40,7,62,42,65,32,82,16,53,86,106,24,118,91,44,14,10,69,88,32,111 110,56,32,114,8,67,110,119,67,67,56,75,29,119,80,104,112,81,113,7,29,114,51,14,90,126 127,40,30,80,27,7,127,122,64,3,115,41,15,7,57,40,79,45,113,10,38,15,29,45,10,109,125 3,73,101,59,14,19,85,105,54,118,123,9,29,53,37,44,37,48,107,44,110,42,27,62,22,9,115,24 123,62,10,99,78,55,72,111,9,55,34,103,107,24,70,77,10

S_72	0,1,2,1,4,77,22,93,8,1,74,65,45,108,127,60,16,16,22,20,68,12,82,24,24,16,94,84,109 45,59,121,32,65,43,72,36,13,63,20,44,69,103,12,9,40,82,113,34,66,45,79,118,94,105 67,46,70,97,11,91,123,4,38,64,81,66,81,64,25,82,9,90,67,24,3,123,42,41,122,120,104 126,108,40,112,62,100,98,122,36,62,19,67,69,23,32,81,43,88,32,25,59,0,62,71,117,14 31,46,68,119,10,122,5,119,90,98,69,127,20,108,91,33,101,85,58,8
S_73	117,103,97,117,17,23,53,53,80,10,100,56,41,103,45,101,51,44,93,68,62,53,96,109,10 93,68,21,26,89,100,33,17,87,109,45,2,80,78,26,91,85,7,15,85,79,57,37,67,8,69,8,57 102,15,86,21,22,51,54,114,101,100,117,54,41,56,33,53,62,11,6,120,47,86,7,102,37,120 61,47,61,91,79,69,67,1,1,125,39,41,117,10,68,110,38,6,77,96,45,114,45,36,125,39,36 97,100,78,89,56,41,11,77,23,87,22,68,58,110,54,56,10,2,54,44,58,38
S_74	0,0,0,59,0,59,111,111,0,7,67,127,85,105,121,126,0,61,111,105,74,76,74,119,22,44 58,59,9,8,74,112,0,30,9,44,74,111,44,50,91,66,17,51,68,102,97,120,91,120,61,37,91 67,82,113,22,50,51,44,67,92,9,45,0,76,62,73,55,64,102,42,112,59,13,125,18,98,0,75 9,120,88,18,116,62,74,59,111,25,125,48,71,10,58,76,38,116,17,120,91,50,3,81,13,88 121,23,37,75,62,107,116,27,44,120,67,23,116,27,73,33,82,1,43,120,95,55
S_75	0,0,0,39,0,45,57,51,0,94,97,24,14,125,86,2,0,99,57,125,100,42,100,13,111,82,55,45,5 21,100,83,0,100,15,76,100,45,82,60,65,123,47,50,43,60,124,76,75,76,125,93,75,97,68 73,101,60,50,76,107,31,5,86,0,42,85,88,80,87,60,28,83,39,103,52,13,84,0,126,5,76 105,7,49,85,100,39,57,46,52,4,3,57,55,42,127,49,37,76,75,40,40,108,109,125,86,97,87 106,85,79,49,28,82,88,97,97,59,28,76,63,78,26,18,76,41,80
S_76	0,0,0,115,0,100,22,1,0,24,57,82,71,59,104,103,0,48,20,87,68,16,70,97,127,87,82,9 124,48,71,120,0,122,120,113,3,29,109,0,89,59,24,9,29,27,74,63,36,110,72,113,99,77 25,68,2,80,87,118,2,52,65,4,0,61,62,112,1,88,41,3,61,24,58,108,123,58,106,88,4,9 46,80,65,40,125,103,70,83,85,51,68,53,65,67,105,46,47,27,107,72,59,107,13,82,114 94,72,115,33,105,73,62,27,31,15,28,75,43,82,61,57,37,83,88,46,86

8-bit S-box implementations

χ_8
A = 8 X = 16

$$
\begin{aligned}
l_0 &= x_7 \oplus x_5;\\
l_1 &= x_6 \oplus x_4;\\
l_2 &= x_5 \oplus x_3;\\
l_3 &= x_4 \oplus x_2;\\
l_4 &= x_7 \oplus x_1;\\
l_5 &= x_3 \oplus x_1;\\
l_6 &= x_6 \oplus x_0;\\
l_7 &= x_2 \oplus x_0;\\
q_0 &= (x_7 \times x_0);\\
q_1 &= (x_1 \times x_2);\\
q_2 &= (x_0 \times x_1);\\
y_0 &= q_1 \oplus l_7;\\
q_3 &= (x_6 \times x_5);\\
y_4 &= q_3 \oplus l_1;\\
q_4 &= (x_4 \times x_3);\\
y_2 &= q_4 \oplus l_3;\\
y_6 &= q_0 \oplus l_6;\\
y_7 &= q_0 \oplus q_2 \oplus l_4;\\
q_5 &= (x_7 \times x_6);\\
y_5 &= q_5 \oplus l_0;\\
q_6 &= (x_3 \times x_2);\\
y_1 &= q_6 \oplus l_5;\\
q_7 &= (x_5 \times x_4);\\
y_3 &= q_7 \oplus l_2;
\end{aligned}
$$

X^3

A= 14 X= 77

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_7 \oplus x_3;\\
l_2 &= x_5 \oplus x_3;\\
l_3 &= l_1 \oplus x_1;\\
l_4 &= x_4 \oplus x_2;\\
l_5 &= l_4 \oplus x_1;\\
l_6 &= x_7 \oplus l_0;\\
l_7 &= l_0 \oplus l_5;\\
l_8 &= l_0 \oplus l_1;\\
l_9 &= l_6 \oplus x_0;\\
l_{10} &= x_7 \oplus x_1;\\
l_{11} &= l_{10} \oplus l_9;\\
l_{12} &= x_6 \oplus l_6;\\
l_{13} &= l_{12} \oplus l_4;\\
l_{14} &= l_1 \oplus l_9;\\
l_{15} &= l_{12} \oplus l_{14};\\
l_{16} &= x_4 \oplus l_{10};\\
l_{17} &= l_{13} \oplus l_{16};\\
l_{18} &= l_1 \oplus x_2;\\
l_{19} &= l_{16} \oplus x_0;\\
l_{20} &= l_{15} \oplus l_{19};\\
l_{21} &= x_4 \oplus l_{18};\\
l_{22} &= x_6 \oplus l_{14};\\
l_{23} &= l_{12} \oplus x_1;\\
l_{24} &= l_6 \oplus l_{20};\\
l_{25} &= x_6 \oplus l_{13};\\
l_{26} &= l_{25} \oplus x_0;\\
l_{27} &= l_{16} \oplus l_{15};\\
l_{28} &= l_{20} \oplus l_5;\\
l_{29} &= l_{25} \oplus l_{28};\\
l_{30} &= x_7 \oplus l_4;\\
l_{31} &= x_4 \oplus l_{28};\\
l_{32} &= l_{29} \oplus l_{14};\\
l_{33} &= x_7 \oplus l_{32};\\
q_0 &= (x_6 \times l_0);\\
q_1 &= (l_{28} \times l_9);\\
l_{38} &= q_0 \oplus q_1;\\
q_2 &= (l_{21} \times l_{15}) \oplus l_1;\\
y_0 &= l_{38} \oplus q_2 \oplus l_{32};\\
q_3 &= (x_3 \times l_{25});\\
q_4 &= (l_6 \times l_3) \oplus x_7;\\
l_{34} &= q_3 \oplus q_4;\\
q_5 &= (l_{18} \times l_{17}) \oplus x_2;\\
l_{35} &= l_{34} \oplus q_5;\\
y_5 &= l_{35} \oplus l_{29};\\
q_6 &= (l_2 \times l_{23}) \oplus x_5;\\
q_7 &= (l_{31} \times l_{19}) \oplus x_4;\\
l_{36} &= q_2 \oplus q_7;\\
y_7 &= l_{35} \oplus q_6 \oplus l_{36} \oplus x_4;\\
q_8 &= (l_5 \times l_{22}) \oplus x_4;\\
q_9 &= (l_{26} \times l_{27}) \oplus x_0;\\
y_6 &= l_{34} \oplus l_{36} \oplus q_8 \oplus q_9 \oplus l_{30};\\
q_{10} &= (l_{11} \times l_{24}) \oplus x_5 \oplus x_1;\\
y_3 &= l_{38} \oplus q_5 \oplus q_3 \oplus q_{10} \oplus q_9 \oplus l_{20};\\
q_{11} &= (l_7 \times l_8) \oplus x_5;\\
y_4 &= l_{38} \oplus q_{11} \oplus q_4 \oplus q_8 \oplus q_9 \oplus l_1;\\
q_{12} &= (l_{16} \times l_{14}) \oplus x_4;\\
l_{37} &= q_{11} \oplus q_{12};\\
y_1 &= l_{38} \oplus q_5 \oplus q_7 \oplus l_{37} \oplus l_2;\\
q_{13} &= (l_{33} \times l_{13}) \oplus x_6;\\
y_2 &= l_{37} \oplus q_{13} \oplus l_{12};
\end{aligned}
$$

X^5

A= 12 X= 71

$$
\begin{aligned}
l_0 &= x_7 \oplus x_4;\\
l_1 &= x_5 \oplus x_3;\\
l_2 &= x_6 \oplus x_2;\\
l_3 &= x_5 \oplus x_1;\\
l_4 &= x_5 \oplus x_0;\\
l_5 &= x_3 \oplus x_0;\\
l_6 &= x_3 \oplus l_2;\\
l_7 &= l_6 \oplus l_3;\\
l_8 &= x_5 \oplus l_0;\\
l_9 &= l_8 \oplus x_2;\\
l_{10} &= l_0 \oplus l_1;\\
l_{11} &= x_7 \oplus x_6;\\
l_{12} &= l_{11} \oplus l_8;\\
l_{13} &= x_3 \oplus l_9;\\
l_{14} &= x_4 \oplus x_1;\\
l_{15} &= l_{14} \oplus x_0;\\
l_{16} &= l_{11} \oplus l_4;\\
l_{17} &= x_1 \oplus l_4;\\
l_{18} &= l_1 \oplus l_{17};\\
l_{19} &= x_7 \oplus l_{13};\\
l_{20} &= x_4 \oplus l_2;\\
l_{21} &= l_6 \oplus l_{14};\\
l_{22} &= l_1 \oplus l_{14};\\
l_{23} &= l_8 \oplus l_{22};\\
l_{24} &= x_2 \oplus l_{21};\\
l_{25} &= l_9 \oplus l_{15};\\
l_{26} &= x_4 \oplus l_{19};\\
l_{27} &= x_5 \oplus l_2;\\
l_{28} &= l_{23} \oplus l_{16};\\
l_{29} &= l_0 \oplus l_{21};\\
l_{30} &= l_{11} \oplus l_{20};\\
l_{31} &= x_5 \oplus l_{13};\\
q_0 &= (l_{20} \times l_{25}) \oplus x_2;\\
q_1 &= (l_{26} \times l_{15});\\
y_2 &= q_0 \oplus q_1 \oplus l_{31};\\
q_2 &= (l_{19} \times l_{24}) \oplus x_4 \oplus x_3;\\
q_3 &= (l_9 \times l_{28}) \oplus x_5;\\
q_4 &= (l_5 \times l_6) \oplus x_3;\\
l_{32} &= q_1 \oplus q_3;\\
l_{33} &= q_2 \oplus q_4;\\
y_3 &= l_{32} \oplus l_{33} \oplus x_7;\\
q_5 &= (x_5 \times l_{23});\\
q_6 &= (l_{27} \times l_{22}) \oplus x_5;\\
l_{34} &= q_5 \oplus q_6;\\
y_5 &= l_{34} \oplus l_{21};\\
y_0 &= l_{34} \oplus l_{33} \oplus l_4;\\
q_7 &= (l_3 \times l_{13}) \oplus x_5;\\
q_8 &= (l_{18} \times l_{10}) \oplus x_3;\\
y_6 &= q_7 \oplus q_8 \oplus q_0 \oplus q_3 \oplus l_2;\\
q_9 &= (l_{16} \times l_7) \oplus x_6 \oplus x_5;\\
l_{35} &= q_1 \oplus q_9;\\
y_1 &= q_5 \oplus q_2 \oplus q_7 \oplus l_{35} \oplus l_1;\\
q_{10} &= (l_{12} \times l_{17}) \oplus x_5;\\
y_7 &= l_{34} \oplus q_2 \oplus q_{10} \oplus l_{32} \oplus l_0;\\
q_{11} &= (l_8 \times l_{29}) \oplus x_7;\\
y_4 &= q_{11} \oplus q_8 \oplus q_4 \oplus l_{35} \oplus l_{30};
\end{aligned}
$$

X^9

A = 14 X = 75

$$
\begin{aligned}
l_0 &= x_7 \oplus x_5;\\
l_1 &= x_6 \oplus x_5;\\
l_2 &= x_7 \oplus x_4;\\
l_3 &= x_6 \oplus x_4;\\
l_4 &= x_5 \oplus x_3;\\
l_5 &= x_6 \oplus x_2;\\
l_6 &= x_6 \oplus x_1;\\
l_7 &= x_3 \oplus x_1;\\
l_8 &= x_2 \oplus x_0;\\
l_9 &= l_1 \oplus l_8;\\
l_{10} &= x_4 \oplus l_4;\\
l_{11} &= l_2 \oplus l_7;\\
l_{12} &= l_{11} \oplus x_0;\\
l_{13} &= l_0 \oplus l_3;\\
l_{14} &= l_{13} \oplus l_{12};\\
l_{15} &= x_5 \oplus l_7;\\
l_{16} &= x_3 \oplus l_{12};\\
l_{17} &= l_2 \oplus l_5;\\
l_{18} &= x_2 \oplus l_7;\\
l_{19} &= l_{17} \oplus l_6;\\
l_{20} &= l_{13} \oplus x_3;\\
l_{21} &= l_{17} \oplus l_8;\\
l_{22} &= x_7 \oplus l_{17};\\
l_{23} &= l_{22} \oplus l_{16};\\
l_{24} &= x_2 \oplus l_6;\\
l_{25} &= x_3 \oplus l_{22};\\
l_{26} &= l_{13} \oplus x_1;\\
l_{27} &= l_0 \oplus l_{12};\\
l_{28} &= x_7 \oplus l_{10};\\
l_{29} &= l_3 \oplus l_9;\\
l_{30} &= x_6 \oplus l_{25};\\
l_{31} &= x_7 \oplus l_{29};\\
q_0 &= (l_5 \times l_{20}) \oplus x_6;\\
q_1 &= (l_{18} \times l_{28}) \oplus x_3;\\
q_2 &= (l_{11} \times l_{21}) \oplus l_2;\\
l_{32} &= q_1 \oplus q_2;\\
y_0 &= q_0 \oplus l_{32} \oplus l_{31};\\
q_3 &= (l_{10} \times l_9) \oplus x_5;\\
q_4 &= (l_7 \times l_8);\\
q_5 &= (l_{15} \times l_{29}) \oplus x_5;\\
l_{33} &= q_2 \oplus q_4;\\
l_{34} &= q_3 \oplus q_5;\\
y_4 &= l_{33} \oplus l_{34} \oplus l_6;\\
q_6 &= (l_2 \times l_5);\\
q_7 &= (l_{13} \times l_{23}) \oplus x_7 \oplus x_6;\\
y_2 &= q_6 \oplus q_7 \oplus q_5 \oplus l_{30};\\
q_8 &= (l_{22} \times l_{14}) \oplus x_6;\\
q_9 &= (l_{25} \times l_{16}) \oplus x_4;\\
l_{35} &= q_8 \oplus q_9;\\
y_7 &= l_{35} \oplus l_{33} \oplus l_0;\\
q_{10} &= (l_{24} \times l_{27}) \oplus x_1;\\
l_{36} &= q_7 \oplus q_{10};\\
y_5 &= l_{34} \oplus q_8 \oplus q_6 \oplus q_{10} \oplus l_{19};\\
q_{11} &= (l_3 \times l_6) \oplus x_6;\\
l_{37} &= l_{36} \oplus q_{11};\\
y_3 &= q_6 \oplus l_{37} \oplus l_{18};\\
q_{12} &= (l_{17} \times l_{12}) \oplus l_2;\\
y_6 &= l_{37} \oplus q_{12} \oplus q_9 \oplus l_{32} \oplus l_1;\\
q_{13} &= (l_4 \times l_{26}) \oplus x_5;\\
y_1 &= q_0 \oplus q_{13} \oplus l_{36} \oplus q_{12} \oplus l_{35} \oplus x_1;
\end{aligned}
$$

X^{17}

A = 10 X = 48

$$
\begin{aligned}
l_0 &= x_6 \oplus x_4;\\
l_1 &= x_7 \oplus x_2;\\
l_2 &= x_3 \oplus x1;\\
l_3 &= x_7 \oplus x0;\\
l_4 &= x_5 \oplus x_3;\\
l_5 &= l_4 \oplus x0;\\
l_6 &= x_5 \oplus l_1;\\
l_7 &= l_6 \oplus x1;\\
l_8 &= x_4 \oplus x_2;\\
l_9 &= l_8 \oplus l_7;\\
l_{10} &= x_6 \oplus x1;\\
l_{11} &= l_{10} \oplus x0;\\
l_{12} &= x_4 \oplus l_{11};\\
l_{13} &= l_{10} \oplus l_9;\\
l_{14} &= l_8 \oplus l_2;\\
l_{15} &= x_7 \oplus x_6;\\
l_{16} &= l_{15} \oplus l_6;\\
l_{17} &= x_4 \oplus l_2;\\
l_{18} &= l_4 \oplus l_8;\\
l_{19} &= x_4 \oplus l_3;\\
l_{20} &= l_{15} \oplus l_2;\\
l_{21} &= x_3 \oplus l_6;\\
l_{22} &= x_6 \oplus l_{18};\\
q_0 &= (l_{18} \times l_{20}) \oplus x_3;\\
q_1 &= (l_2 \times l_{22}) \oplus x_3;\\
q_2 &= (l_0 \times l_3);\\
q_3 &= (l_6 \times l_5) \oplus x_5;\\
y_0 &= q_0 \oplus q_1 \oplus q_2 \oplus q_3 \oplus l_{19};\\
q_4 &= (x_7 \times x_6);\\
q_5 &= (x_3 \times x_2);\\
q_6 &= (l_3 \times l_{17});\\
l_{23} &= q_0 \oplus q_4;\\
l_{24} &= l_{23} \oplus q_6;\\
y_1 &= l_{24} \oplus q_5 \oplus l_{21};\\
q_7 &= (l_{13} \times l_7) \oplus x_7 \oplus x_5;\\
q_8 &= (l_{16} \times l_{12}) \oplus x_6;\\
y_4 &= l_{23} \oplus q_7 \oplus q_8 \oplus l_9;\\
y_3 &= q_5 \oplus q_7 \oplus q_8 \oplus q_6 \oplus l_{14};\\
q_9 &= (l_1 \times l_{11});\\
y_2 &= l_{24} \oplus q_9 \oplus q_3 \oplus q_7 \oplus l_{16};\\
y_6 &= y_2;\\
y_7 &= y_4;\\
y_5 &= 0;
\end{aligned}
$$

X^{33}

A = 14 X = 76

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_7 \oplus x_3;\\
l_2 &= x_5 \oplus x_3;\\
l_3 &= x_7 \oplus x_2;\\
l_4 &= x_5 \oplus x_2;\\
l_5 &= x_5 \oplus x_1;\\
l_6 &= x_3 \oplus x_1;\\
l_7 &= x_6 \oplus x_0;\\
l_8 &= l_2 \oplus x_0;\\
l_9 &= x_4 \oplus l_3;\\
l_{10} &= l_9 \oplus x_1;\\
l_{11} &= x_7 \oplus l_8;\\
l_{12} &= l_7 \oplus l_{11};\\
l_{13} &= x_4 \oplus l_4;\\
l_{14} &= l_4 \oplus l_7;\\
l_{15} &= l_{10} \oplus l_7;\\
l_{16} &= x_7 \oplus l_0;\\
l_{17} &= l_{12} \oplus x_2;\\
l_{18} &= l_0 \oplus l_8;\\
l_{19} &= x_3 \oplus l_{12};\\
l_{20} &= l_0 \oplus l_{15};\\
l_{21} &= x_6 \oplus l_6;\\
l_{22} &= x_3 \oplus l_{14};\\
l_{23} &= x_7 \oplus l_{21};\\
l_{24} &= x_5 \oplus l_{10};\\
l_{25} &= l_1 \oplus l_{18};\\
l_{26} &= x_4 \oplus l_2;\\
l_{27} &= x_7 \oplus l_{26};\\
l_{28} &= x_6 \oplus l_{10};\\
l_{29} &= x_5 \oplus l_{27};\\
l_{30} &= x_1 \oplus l_{18};\\
l_{31} &= l_{26} \oplus l_{23};\\
l_{32} &= l_9 \oplus l_{20};\\
q_0 &= (l_0 \times l_2) \oplus x_5;\\
q_1 &= (l_{18} \times l_{17}) \oplus x_3;\\
q_2 &= (l_{30} \times l_{31}) \oplus x_4 \oplus x_1;\\
l_{33} &= q_0 \oplus q_2;\\
l_{34} &= l_{33} \oplus q_1;\\
y_0 &= l_{34} \oplus l_{20};\\
q_3 &= (l_{13} \times l_{12}) \oplus x_5;\\
q_4 &= (l_{26} \times l_{14}) \oplus x_5;\\
q_5 &= (l_6 \times l_7);\\
l_{35} &= q_4 \oplus q_5;\\
l_{36} &= q_3 \oplus l_{35};\\
y_2 &= l_{36} \oplus l_{23};\\
q_6 &= (l_{19} \times l_{29}) \oplus x_7;\\
q_7 &= (l_4 \times l_{32}) \oplus x_5;\\
l_{37} &= q_1 \oplus q_7;\\
y_7 &= l_{36} \oplus l_{37} \oplus q_6 \oplus l_{27};\\
q_8 &= (l_1 \times l_{15}) \oplus x_7;\\
y_6 &= l_{33} \oplus l_{35} \oplus q_8 \oplus q_7 \oplus l_3;\\
q_9 &= (l_{21} \times l_{24}) \oplus x_1;\\
q_{10} &= (l_9 \times l_{22}) \oplus x_2;\\
l_{38} &= q_6 \oplus q_{10};\\
y_4 &= l_{38} \oplus q_9 \oplus q_5 \oplus l_{16};\\
q_{11} &= (x_2 \times l_8);\\
y_3 &= l_{34} \oplus q_{11} \oplus l_{38} \oplus q_8 \oplus x_4;\\
q_{12} &= (l_5 \times l_{25});\\
l_{39} &= q_{11} \oplus q_{12};\\
y_1 &= l_{33} \oplus l_{35} \oplus l_{39} \oplus q_6 \oplus l_{21};\\
q_{13} &= (l_{11} \times l_{28}) \oplus x_7;\\
y_5 &= l_{39} \oplus l_{37} \oplus q_4 \oplus q_{13} \oplus l_{10};
\end{aligned}
$$

X^{65}

A= 12 X= 68

$$
\begin{aligned}
l_0 &= x_7 \oplus x_3;\\
l_1 &= x_5 \oplus x_3;\\
l_2 &= x_6 \oplus x_1;\\
l_3 &= x_3 \oplus x_1;\\
l_4 &= x_6 \oplus x_0;\\
l_5 &= x_4 \oplus l_3;\\
l_6 &= x_6 \oplus x_2;\\
l_7 &= l_6 \oplus l_5;\\
l_8 &= x_5 \oplus l_2;\\
l_9 &= x_7 \oplus l_2;\\
l_{10} &= l_8 \oplus l_7;\\
l_{11} &= l_{10} \oplus x_0;\\
l_{12} &= l_6 \oplus l_9;\\
l_{13} &= l_{12} \oplus l_{11};\\
l_{14} &= l_0 \oplus l_4;\\
l_{15} &= x_5 \oplus l_{12};\\
l_{16} &= x_7 \oplus l_3;\\
l_{17} &= l_3 \oplus l_4;\\
l_{18} &= l_{14} \oplus l_{11};\\
l_{19} &= x_7 \oplus l_{15};\\
l_{20} &= x_3 \oplus l_{15};\\
l_{21} &= x_6 \oplus l_{15};\\
l_{22} &= x_2 \oplus l_{11};\\
l_{23} &= l_1 \oplus l_{16};\\
l_{24} &= l_1 \oplus l_6;\\
l_{25} &= l_{24} \oplus l_{22};\\
l_{26} &= x_7 \oplus l_{14};\\
l_{27} &= l_8 \oplus l_{16};\\
l_{28} &= x_3 \oplus l_{10};\\
l_{29} &= l_6 \oplus l_{18};\\
q_0 &= (l_0 \times l_{12}) \oplus x_7;\\
q_1 &= (l_8 \times l_{11}) \oplus x_5;\\
y_2 &= q_0 \oplus q_1 \oplus l_{20};\\
q_2 &= (l_3 \times l_{26}) \oplus x_3;\\
q_3 &= (l_{17} \times l_{20}) \oplus l_3;\\
l_{30} &= q_1 \oplus q_2;\\
y_0 &= q_0 \oplus l_{30} \oplus q_3 \oplus l_4;\\
q_4 &= (l_{29} \times l_7) \oplus x_4;\\
q_5 &= (l_{16} \times l_{18}) \oplus x_7;\\
y_5 &= q_4 \oplus q_5 \oplus l_7;\\
q_6 &= (l_{27} \times l_{25}) \oplus x_6;\\
q_7 &= (l_{28} \times l_{14});\\
y_1 &= q_6 \oplus q_7 \oplus q_2 \oplus q_3 \oplus l_{19};\\
q_8 &= (l_{24} \times l_{13}) \oplus l_1;\\
l_{31} &= q_5 \oplus q_8;\\
y_3 &= q_4 \oplus l_{31} \oplus l_{30} \oplus q_3 \oplus l_{21};\\
q_9 &= (l_9 \times l_{22});\\
y_4 &= q_0 \oplus l_{31} \oplus q_7 \oplus q_9 \oplus l_{30} \oplus l_{23};\\
q_{10} &= (x_5 \times l_5);\\
y_6 &= q_{10} \oplus q_6 \oplus q_9 \oplus l_{30} \oplus l_1;\\
q_{11} &= (l_4 \times l_{15});\\
y_7 &= q_{10} \oplus q_4 \oplus q_2 \oplus q_{11} \oplus l_2;
\end{aligned}
$$

X^{129}

A= 14 X= 81

$$
\begin{aligned}
l_0 &= x_5 \oplus x_3;\\
l_1 &= x_6 \oplus x_1;\\
l_2 &= x_5 \oplus x_1;\\
l_3 &= x_3 \oplus x_0;\\
l_4 &= l_2 \oplus l_3;\\
l_5 &= x_7 \oplus l_1;\\
l_6 &= l_5 \oplus l_4;\\
l_7 &= x_4 \oplus l_2;\\
l_8 &= x_4 \oplus l_0;\\
l_9 &= l_8 \oplus l_6;\\
l_{10} &= x_2 \oplus l_5;\\
l_{11} &= l_{10} \oplus x_0;\\
l_{12} &= x_3 \oplus l_1;\\
l_{13} &= x_0 \oplus l_6;\\
l_{14} &= l_{13} \oplus l_4;\\
l_{15} &= l_7 \oplus x_0;\\
l_{16} &= x_5 \oplus l_4;\\
l_{17} &= x_7 \oplus l_8;\\
l_{18} &= l_{17} \oplus l_3;\\
l_{19} &= x_2 \oplus l_2;\\
l_{20} &= l_{12} \oplus l_{15};\\
l_{21} &= l_{12} \oplus l_{10};\\
l_{22} &= l_{18} \oplus l_6;\\
l_{23} &= l_{12} \oplus l_6;\\
l_{24} &= l_9 \oplus l_{11};\\
l_{25} &= l_{12} \oplus l_{19};\\
l_{26} &= x_3 \oplus l_{20};\\
l_{27} &= l_8 \oplus l_{21};\\
l_{28} &= l_0 \oplus l_5;\\
l_{29} &= x_2 \oplus l_{16};\\
l_{30} &= x_3 \oplus l_{27};\\
l_{31} &= x_4 \oplus l_{10};\\
l_{32} &= x_5 \oplus l_{15};\\
l_{33} &= x_3 \oplus l_{24};\\
q_0 &= (l_3 \times l_5);\\
q_1 &= (l_2 \times l_{26}) \oplus x_5;\\
q_2 &= (l_{15} \times l_{28}) \oplus l_2;\\
l_{34} &= q_1 \oplus q_2;\\
l_{35} &= l_{34} \oplus q_0;\\
y_0 &= l_{35} \oplus l_{23};\\
q_3 &= (x_6 \times l_4);\\
q_4 &= (l_{27} \times l_9) \oplus x_7 \oplus x_4;\\
q_5 &= (l_{24} \times l_{20}) \oplus x_4;\\
l_{36} &= q_4 \oplus q_5;\\
y_6 &= q_3 \oplus q_1 \oplus l_{36} \oplus x_4;\\
q_6 &= (l_{16} \times l_{30}) \oplus x_3;\\
q_7 &= (l_{33} \times l_{14}) \oplus x_1;\\
l_{37} &= q_6 \oplus q_7;\\
y_7 &= q_3 \oplus l_{36} \oplus q_0 \oplus l_{37} \oplus l_0;\\
q_8 &= (l_{19} \times l_{17}) \oplus x_5;\\
q_9 &= (l_{21} \times l_{32});\\
y_1 &= l_{36} \oplus q_9 \oplus q_8 \oplus l_1;\\
y_2 &= l_{37} \oplus l_{35} \oplus q_5 \oplus q_8 \oplus x_1;\\
q_{10} &= (l_{22} \times l_5) \oplus x_6;\\
q_{11} &= (l_{18} \times l_{29}) \oplus x_0;\\
y_4 &= q_{10} \oplus q_4 \oplus q_9 \oplus q_7 \oplus q_{11} \oplus l_{12};\\
q_{12} &= (l_7 \times l_{11}) \oplus x_1;\\
y_3 &= q_{10} \oplus q_9 \oplus q_6 \oplus q_{12} \oplus q_2 \oplus l_{25};\\
q_{13} &= (l_{31} \times l_6) \oplus x_7 \oplus x_6;\\
y_5 &= q_8 \oplus l_{36} \oplus l_{34} \oplus q_6 \oplus q_{13} \oplus l_{13};
\end{aligned}
$$

S_A81

A = 14 X = 68

$$
\begin{aligned}
l_0 &= x_7 \oplus x_5;\\
l_1 &= x_5 \oplus x_2;\\
l_2 &= x_2 \oplus x_0;\\
l_3 &= x_6 \oplus x_5;\\
l_4 &= l_3 \oplus x_1;\\
l_5 &= x_3 \oplus l_1;\\
l_6 &= l_5 \oplus l_4;\\
l_7 &= x_7 \oplus l_4;\\
l_8 &= l_5 \oplus x_0;\\
l_9 &= x_3 \oplus l_7;\\
l_{10} &= x_4 \oplus x_1;\\
l_{11} &= l_{10} \oplus l_9;\\
l_{12} &= x_6 \oplus l_{10};\\
l_{13} &= x_7 \oplus l_{12};\\
l_{14} &= x_7 \oplus l_3;\\
l_{15} &= l_{14} \oplus l_8;\\
l_{16} &= l_0 \oplus x_0;\\
l_{17} &= l_3 \oplus l_{10};\\
l_{18} &= l_3 \oplus x_4;\\
l_{19} &= x_3 \oplus l_{16};\\
l_{20} &= l_{17} \oplus l_2;\\
l_{21} &= x_3 \oplus l_{15};\\
l_{22} &= l_{10} \oplus l_{13};\\
l_{23} &= l_3 \oplus l_5;\\
l_{24} &= l_{17} \oplus x_0;\\
l_{25} &= x_2 \oplus l_{10};\\
l_{26} &= x_7 \oplus l_{24};\\
l_{27} &= l_4 \oplus l_{21};\\
l_{28} &= l_{12} \oplus l_{21};\\
l_{29} &= l_5 \oplus l_{12};\\
q_0 &= (l_{10} \times l_{28}) \oplus x_7 \oplus x_4;\\
q_1 &= (l_{13} \times l_{21}) \oplus x_7 \oplus x_6;\\
q_2 &= (l_9 \times l_{20}) \oplus x_7 \oplus x_3;\\
y_7 &= q_0 \oplus q_1 \oplus q_2;\\
q_3 &= (x_4 \times l_{23});\\
q_4 &= (l_{17} \times l_{27}) \oplus x_7;\\
l_{30} &= q_0 \oplus q_4;\\
y_6 &= q_3 \oplus l_{30};\\
q_5 &= (l_4 \times l_{16}) \oplus x_5;\\
q_6 &= (l_{12} \times l_{24}) \oplus x_4;\\
y_5 &= q_3 \oplus q_5 \oplus q_6;\\
q_7 &= (x_2 \times l_{26});\\
q_8 &= (l_{19} \times l_{29}) \oplus x_5 \oplus x_3;\\
l_{31} &= q_3 \oplus q_8;\\
y_4 &= l_{31} \oplus q_5 \oplus q_7;\\
q_9 &= (l_{22} \times l_{25}) \oplus x_1;\\
q_{10} &= (l_0 \times l_{15}) \oplus x_7;\\
l_{32} &= q_1 \oplus q_9;\\
y_3 &= l_{31} \oplus q_{10} \oplus l_{32};\\
q_{11} &= (l_6 \times l_8) \oplus x_3 \oplus x_2;\\
y_1 &= l_{31} \oplus l_{30} \oplus q_{11} \oplus l_{32};\\
q_{12} &= (l_1 \times l_{18}) \oplus x_5;\\
y_2 &= q_{12} \oplus q_{10} \oplus l_{30} \oplus q_5 \oplus q_2 \oplus q_{11};\\
q_{13} &= (l_2 \times l_7);\\
y_0 &= q_9 \oplus q_5 \oplus q_{13} \oplus q_8 \oplus x_0;
\end{aligned}
$$

S_81
A = 14 X = 70

$$
\begin{aligned}
l_0 &= x_7 \oplus x_5;\\
l_1 &= x_6 \oplus x_5;\\
l_2 &= x_7 \oplus x_3;\\
l_3 &= x_6 \oplus x_3;\\
l_4 &= x_5 \oplus x_2;\\
l_5 &= l_4 \oplus x_0;\\
l_6 &= l_4 \oplus x_1;\\
l_7 &= l_2 \oplus x_2;\\
l_8 &= x_4 \oplus l_4;\\
l_9 &= x_4 \oplus l_2;\\
l_{10} &= l_9 \oplus x_1;\\
l_{11} &= l_1 \oplus l_5;\\
l_{12} &= l_3 \oplus l_{10};\\
l_{13} &= l_{12} \oplus l_{11};\\
l_{14} &= l_7 \oplus l_5;\\
l_{15} &= l_6 \oplus l_{13};\\
l_{16} &= x_6 \oplus l_7;\\
l_{17} &= l_{16} \oplus l_{13};\\
l_{18} &= l_3 \oplus l_{13};\\
l_{19} &= x_1 \oplus l_{17};\\
l_{20} &= x_6 \oplus l_9;\\
l_{21} &= x_5 \oplus l_{15};\\
l_{22} &= l_3 \oplus l_6;\\
l_{23} &= l_1 \oplus x_1;\\
l_{24} &= l_1 \oplus l_{22};\\
l_{25} &= l_{20} \oplus l_{22};\\
l_{26} &= x_7 \oplus l_4;\\
l_{27} &= l_{20} \oplus l_{14};\\
l_{28} &= l_3 \oplus l_{23};\\
l_{29} &= l_{26} \oplus l_{15};\\
l_{30} &= x_7 \oplus l_{12};\\
l_{31} &= x_4 \oplus l_{17};\\
q_0 &= (x_7 \times x_3);\\
q_1 &= (l_5 \times l_{20});\\
q_2 &= (l_{28} \times l_{13}) \oplus x_1;\\
l_{32} &= q_1 \oplus q_2;\\
y_0 &= q_0 \oplus l_{32} \oplus l_{12};\\
q_3 &= (l_1 \times l_7);\\
q_4 &= (l_2 \times l_8);\\
q_5 &= (l_{19} \times l_{25}) \oplus x_4;\\
y_1 &= q_3 \oplus q_4 \oplus q_2 \oplus q_5 \oplus l_{27};\\
q_6 &= (l_9 \times l_{26}) \oplus x_7;\\
q_7 &= (l_{23} \times l_{29});\\
l_{33} &= q_2 \oplus q_6;\\
l_{34} &= l_{33} \oplus q_7;\\
y_3 &= l_{34} \oplus l_6;\\
q_8 &= (l_0 \times l_{11});\\
q_9 &= (x_2 \times l_{17});\\
q_{10} &= (l_{15} \times l_{31}) \oplus x_0;\\
y_6 &= q_8 \oplus q_9 \oplus q_2 \oplus q_{10} \oplus l_{26};\\
q_{11} &= (l_{16} \times l_{10}) \oplus l_2;\\
l_{35} &= q_7 \oplus q_{11};\\
y_5 &= q_0 \oplus q_4 \oplus l_{35} \oplus l_{24};\\
q_{12} &= (l_0 \times l_{18}) \oplus x_7;\\
y_4 &= l_{35} \oplus q_{12} \oplus l_{32} \oplus q_3 \oplus q_9 \oplus l_{30};\\
y_7 &= q_6 \oplus q_1 \oplus q_8 \oplus l_{35} \oplus q_5 \oplus l_3;\\
q_{13} &= (l_{21} \times l_{22});\\
y_2 &= q_0 \oplus q_3 \oplus l_{34} \oplus q_{13} \oplus l_{14};
\end{aligned}
$$

S_82
A = 11 X = 48

$$
\begin{aligned}
l_0 &= x_5 \oplus x_4;\\
l_1 &= x_7 \oplus x_2;\\
l_2 &= x_3 \oplus x_1;\\
l_3 &= x_4 \oplus x_1;\\
l_4 &= l_3 \oplus x_0;\\
l_5 &= x_6 \oplus x_2;\\
l_6 &= l_5 \oplus l_2;\\
l_7 &= x_5 \oplus l_3;\\
l_8 &= x_7 \oplus l_0;\\
l_9 &= l_8 \oplus x_0;\\
l_{10} &= x_3 \oplus l_9;\\
l_{11} &= x_7 \oplus x_5;\\
l_{12} &= l_{11} \oplus x_3;\\
l_{13} &= l_1 \oplus l_9;\\
l_{14} &= l_5 \oplus x_0;\\
l_{15} &= x_6 \oplus l_3;\\
l_{16} &= x_7 \oplus l_{12};\\
l_{17} &= l_{16} \oplus l_5;\\
l_{18} &= x_3 \oplus l_6;\\
l_{19} &= l_{11} \oplus l_{15};\\
l_{20} &= l_{16} \oplus l_{13};\\
l_{21} &= x_4 \oplus l_1;\\
l_{22} &= x_5 \oplus l_{10};\\
l_{23} &= l_{16} \oplus l_{14};\\
q_0 &= (l_7 \times l_{17}) \oplus x_5;\\
q_1 &= (l_{21} \times l_{23}) \oplus x_2;\\
y_1 &= q_0 \oplus q_1;\\
q_2 &= (l_8 \times l_{18});\\
q_3 &= (l_{20} \times l_{19}) \oplus x_4;\\
q_4 &= (l_4 \times l_6) \oplus x_1;\\
y_0 &= q_2 \oplus q_0 \oplus q_3 \oplus q_4;\\
q_5 &= (l_{14} \times l_9) \oplus x_0;\\
q_6 &= (l_2 \times l_{22}) \oplus x_3;\\
y_2 &= q_0 \oplus q_5 \oplus q_6;\\
q_7 &= (l_0 \times l_1);\\
q_8 &= (l_{12} \times l_{13}) \oplus x_5;\\
y_5 &= q_7 \oplus q_8 \oplus q_1;\\
y_3 &= q_7 \oplus q_5;\\
q_9 &= (l_{15} \times l_{10}) \oplus x_4;\\
y_6 &= q_8 \oplus q_9 \oplus q_3;\\
q_{10} &= (x_4 \times x_3);\\
y_4 &= q_{10} \oplus q_3 \oplus q_6;\\
y_7 &= q_{10} \oplus q_2 \oplus q_1 \oplus q_9;
\end{aligned}
$$

S_83
A = 10 X = 46

$$
\begin{aligned}
l_0 &= x_5 \oplus x_1;\\
l_1 &= x_4 \oplus x_0;\\
l_2 &= x_3 \oplus x_0;\\
l_3 &= x_7 \oplus l_2;\\
l_4 &= x_7 \oplus x_5;\\
l_5 &= l_4 \oplus l_1;\\
l_{20} &= x_4 \oplus x_6;\\
l_{21} &= x_5 \oplus x_3;\\
l_6 &= l_{20} \oplus l_{21};\\
l_7 &= x_6 \oplus l_5;\\
l_8 &= l_4 \oplus x_2;\\
l_9 &= x_2 \oplus l_0;\\
l_{10} &= x_5 \oplus l_7;\\
l_{22} &= l_0 \oplus l_{20};\\
l_{23} &= x_2 \oplus x_7;\\
l_{11} &= l_{22} \oplus l_{23};\\
l_{12} &= l_9 \oplus x_0;\\
l_{13} &= x_3 \oplus l_6;\\
l_{14} &= l_5 \oplus l_{12};\\
l_{15} &= x_4 \oplus l_3;\\
l_{16} &= l_6 \oplus x_1;\\
l_{17} &= l_8 \oplus l_7;\\
l_{18} &= l_{13} \oplus l_1;\\
l_{19} &= l_{14} \oplus l_{15};\\
q_0 &= (x_3 \times l_{10});\\
q_1 &= (l_2 \times l_{11});\\
y_2 &= q_0 \oplus q_1;\\
q_2 &= (l_0 \times l_5) \oplus x_5;\\
q_3 &= (l_1 \times l_{16}) \oplus x_4;\\
l_{24} &= q_2 \oplus q_3;\\
q_4 &= (l_9 \times l_{15});\\
l_{25} &= l_{24} \oplus q_4;\\
y_0 &= q_0 \oplus l_{25};\\
q_5 &= (l_{14} \times l_{17}) \oplus x_4 \oplus x_2;\\
q_6 &= (l_{18} \times l_{19}) \oplus x_0;\\
y_4 &= q_5 \oplus q_6;\\
q_7 &= (x_1 \times l_3);\\
y_3 &= q_7 \oplus q_4 \oplus q_1;\\
q_8 &= (l_{13} \times l_8) \oplus x_5;\\
q_9 &= (x_0 \times l_6);\\
l_{26} &= q_7 \oplus q_9;\\
y_1 &= q_8 \oplus l_{26} \oplus q_2 \oplus q_5;\\
y_7 &= q_8 \oplus l_{24} \oplus q_1 \oplus q_6;\\
y_6 &= q_0 \oplus l_{26} \oplus q_6;\\
y_5 &= l_{25} \oplus q_5;
\end{aligned}
$$

X^3
truncated
A= 14 X= 70

$$
\begin{aligned}
l_0 &= x_7 \oplus x_4; & q_2 &= (l_0 \times l_{11}) \oplus x_7;\\
l_1 &= x_6 \oplus x_4; & l_{29} &= q_1 \oplus q_2;\\
l_2 &= x_3 \oplus x_2; & y_0 &= q_0 \oplus l_{29} \oplus l_5;\\
l_3 &= x_6 \oplus x_0; & q_3 &= (l_1 \times l_7);\\
l_4 &= x_2 \oplus x_0; & q_4 &= (l_3 \times l_{25});\\
l_5 &= x_3 \oplus l_3; & l_{30} &= q_0 \oplus q_3;\\
l_6 &= x_5 \oplus l_2; & y_1 &= q_3 \oplus q_2 \oplus q_4 \oplus x_5;\\
l_7 &= l_6 \oplus x_1; & q_5 &= (l_{19} \times l_{16}) \oplus x_7;\\
l_8 &= l_0 \oplus x_1; & q_6 &= (l_{18} \times l_{12}) \oplus x_0;\\
l_9 &= l_8 \oplus x_0; & l_{31} &= q_4 \oplus q_5;\\
l_{10} &= x_5 \oplus l_1; & l_{32} &= l_{31} \oplus q_6;\\
l_{11} &= l_{10} \oplus l_9; & y_6 &= l_{32} \oplus l_{17};\\
l_{12} &= x_6 \oplus l_9; & q_7 &= (l_{13} \times l_{14}) \oplus x_5 \oplus x_3;\\
l_{13} &= x_6 \oplus l_6; & q_8 &= (l_4 \times l_{26});\\
l_{14} &= l_5 \oplus l_{11}; & y_7 &= q_0 \oplus q_7 \oplus q_4 \oplus q_8 \oplus x_4;\\
l_{15} &= x_1 \oplus l_{14}; & q_9 &= (l_9 \times l_{21}) \oplus l_0;\\
l_{16} &= l_{10} \oplus l_{15}; & q_{10} &= (l_{23} \times l_{28}) \oplus x_0;\\
l_{17} &= l_{15} \oplus l_2; & y_2 &= q_3 \oplus q_5 \oplus q_9 \oplus q_{10} \oplus x_7;\\
l_{18} &= l_6 \oplus l_4; & q_{11} &= (l_2 \times l_{22}) \oplus x_3;\\
l_{19} &= x_3 \oplus l_{14}; & y_4 &= l_{30} \oplus q_{11} \oplus q_7 \oplus q_2 \oplus q_6 \oplus q_{10} \oplus x_3;\\
l_{20} &= l_{17} \oplus l_8; & q_{12} &= (l_{24} \times l_{27}) \oplus x_5;\\
l_{21} &= x_5 \oplus l_{16}; & y_3 &= l_{30} \oplus q_{12} \oplus l_{32} \oplus l_8;\\
l_{22} &= x_6 \oplus l_{21}; & q_{13} &= (l_{15} \times l_{20}) \oplus x_5;\\
l_{23} &= x_4 \oplus l_3; & y_5 &= q_{13} \oplus q_5 \oplus q_7 \oplus l_{29} \oplus x_5;\\
l_{24} &= l_0 \oplus l_{10};\\
l_{25} &= x_5 \oplus l_8;\\
l_{26} &= l_{24} \oplus l_8;\\
l_{27} &= l_0 \oplus l_{14};\\
l_{28} &= l_{14} \oplus l_4;\\
q_0 &= (x_2 \times l_{10});\\
q_1 &= (x_4 \times l_4);
\end{aligned}
$$

S-box	LUT
χ_8	0,65,130,67,5,68,134,71,10,75,136,73,13,76,142,79,20,85,150,87,17,80,146,83,26 91,152,89,29,92,158,95,40,105,170,107,45,108,174,111,34,99,160,97,37,100,166,103,52 117,182,119,49,112,178,115,58,123,184,121,61,124,190,127,80,17,210,19,85,20,214,23 90,27,216,25,93,28,222,31,68,5,198,7,65,0,194,3,74,11,200,9,77,12,206,15,104,41,234 43,109,44,238,47,98,35,224,33,101,36,230,39,116,53,246,55,113,48,242,51,122,59,248 57,125,60,254,63,160,161,34,163,165,164,38,167,170,171,40,169,173,172,46,175,180 181,54,183,177,176,50,179,186,187,56,185,189,188,62,191,136,137,10,139,141,140,14 143,130,131,0,129,133,132,6,135,148,149,22,151,145,144,18,147,154,155,24,153,157 156,30,159,208,209,82,211,213,212,86,215,218,219,88,217,221,220,94,223,196,197,70 199,193,192,66,195,202,203,72,201,205,204,78,207,232,233,106,235,237,236,110,239 226,227,96,225,229,228,102,231,244,245,118,247,241,240,114,243,250,251,120,249,253 252,126,255
X^3	0,1,8,15,64,85,120,107,58,115,146,221,231,186,127,36,205,193,191,181,228,252,166 184,107,47,185,251,223,143,61,107,38,115,70,21,145,208,193,134,115,110,179,168,89 80,169,166,127,39,101,59,161,237,139,193,182,166,12,26,245,241,127,125,45,161,191 53,10,146,168,54,252,56,206,12,70,150,68,146,191,62,87,208,241,100,41,186,242,59 186,117,33,252,89,130,223,7,37,251,15,195,197,15,97,241,59,173,44,168,70,196,217 12,89,138,96,161,208,23,251,102,219,64,223,86,207,64,117,231,97,245,145,23,181,53 80,138,228,56,41,231,173,101,179,44,221,68,62,181,96,237,10,221,196,21,26,217,228 33,145,87,237,45,130,80,206,26,219,85,7,143,85,207,185,37,195,8,197,8,185,102,143 86,21,150,179,54,242,101,100,245,182,169,56,33,53,62,139,134,120,47,86,7,102,37 120,61,47,61,219,207,197,195,1,1,125,39,41,117,10,68,110,38,134,205,96,45,242,173 36,125,39,36,97,100,206,217,184,169,139,205,23,87,150,196,58,110,182,184,138,130 54,44,58,38
X^5	0,1,32,51,116,108,46,36,38,226,1,215,169,116,244,59,180,233,17,94,32,100,255,169 132,28,38,172,235,106,51,160,3,150,108,235,26,150,15,145,116,36,28,94,150,223,132 223,77,132,167,124,180,100,36,230,44,32,193,223,46,59,185,190,96,174,55,235,1,214 44,233,103,108,55,46,253,239,215,215,38,180,244,116,167,44,15,150,55,96,226,167,77 3,226,190,85,15,77,5,89,26,59,106,3,156,28,145,244,114,145,5,233,239,116,96,5,26 226,239,169,106,51,226,190,100,94,150,156,5,100,239,174,46,44,190,32,124,223,145 233,172,108,59,124,185,1,214,174,114,169,103,214,214,172,190,255,230,255,244,180 185,3,28,235,255,38,32,89,145,233,51,253,44,55,244,174,255,156,223,17,89,89,3,85 193,96,230,17,156,94,193,114,36,253,185,85,26,160,253,239,124,103,230,51,185,193 89,96,106,106,114,167,180,215,214,235,36,230,59,215,1,160,100,108,174,172,124,38 253,156,85,160,167,103,114,17,15,172,160,132,26,193,77,46,169,17,132,94,5,28,85 15,77,55,103
X^9	0,1,58,36,45,100,251,173,12,138,221,68,125,179,64,145,37,169,245,102,242,54,206,21 173,166,150,130,38,101,241,173,193,138,168,252,44,47,169,181,138,70,8,219,59,191,85 206,64,134,195,26,87,217,56,169,143,206,231,185,196,205,64,86,80,87,245,237,146,221 219,139,54,182,120,231,168,96,10,221,245,127,186,47,205,15,110,179,23,26,179,161 115,54,59,97,38,107,208,130,36,33,62,36,7,205,26,207,89,219,168,53,39,231,59,228 223,87,47,184,130,197,117,45,38,41,61,45,161,125,7,196,44,184,102,237,191,228,242 182,110,125,207,195,8,89,68,10,127,102,223,217,146,68,53,252,185,39,242,115,44,186 217,80,97,191,120,185,117,100,107,101,100,61,150,208,33,58,62,58,150,197,101,41,252 96,8,139,23,195,15,196,143,85,182,115,237,127,56,181,251,166,41,107,197,208,251,241 166,241,117,61,62,33,1,1,86,134,110,161,146,10,70,193,181,37,223,80,23,207,145,86 134,145,7,15,120,39,21,85,56,37,184,186,96,53,12,70,143,21,228,97,139,89,12,193
X^{17}	0,1,152,215,78,214,220,10,10,69,68,69,221,11,153,1,153,215,11,11,147,68,11,146,69 69,1,79,214,79,152,79,214,10,220,78,1,68,1,10,79,221,147,79,1,10,215,146,11,152,11 214,152,146,146,214,68,153,146,1,78,10,146,152,68,220,153,79,221,220,10,69,152,78 147,11,152,215,153,152,146,69,69,220,79,1,146,146,152,1,153,78,220,220,215,153,1 68,78,69,1,221,68,214,78,69,215,146,215,69,68,152,147,153,214,146,215,68,152,69,10 78,153,147,215,10,78,221,147,147,221,147,214,79,146,69,69,11,221,221,153,78,11,146 78,1,10,11,79,153,1,153,78,79,220,147,214,78,78,152,215,10,11,152,11,79,221,215 146,1,152,69,215,221,215,147,78,220,152,68,214,221,10,79,221,1,221,79,220,153,214 221,152,1,147,68,10,10,11,69,152,79,69,220,147,221,68,68,10,220,11,147,220,147 215,214,220,68,11,221,147,146,78,1,79,11,214,220,68,153,215,68,220,214,147,215,78 221,11,214,153,146,10,79,214,68,79,147,220,153,153,146,10,214,69,215

S-box	LUT
X^{33}	0,1,39,185,97,59,182,115,134,86,146,221,102,237,130,150,184,44,173,166,110,161,139 219,115,54,85,143,36,58,242,115,145,86,45,117,96,252,44,47,86,64,217,80,38,107,89 139,130,208,12,193,196,205,186,44,8,139,181,169,207,23,130,197,7,196,173,241,10 146,80,87,161,179,56,181,45,100,68,146,173,251,53,252,23,26,127,237,70,193,237 245,125,161,38,101,36,33,21,143,185,231,120,185,195,23,193,138,223,80,45,61,37,181 38,41,15,196,252,168,143,206,191,97,36,62,228,97,245,102,195,207,96,168,166,241 107,41,110,179,127,102,138,12,217,223,221,68,251,166,15,205,10,221,61,117,169,37 110,125,96,53,205,7,101,107,56,169,191,59,33,58,59,228,85,21,231,39,120,39,85,206 58,62,117,100,217,87,70,12,26,207,8,89,179,125,241,251,186,47,182,54,62,33,206,21 182,242,54,242,191,228,120,231,1,1,197,208,127,245,10,68,64,145,47,184,15,7,70 138,150,197,208,150,195,26,56,37,219,89,186,184,168,53,100,61,134,64,8,219,41,101 87,223,134,145
X^{65}	0,1,190,226,46,174,230,59,100,239,1,215,36,46,55,96,32,51,3,77,190,44,235,36,156,5 100,160,108,116,226,167,94,193,174,108,223,193,89,26,46,59,5,77,193,85,156,85,17 156,124,172,32,44,59,106,185,190,15,85,230,96,38,253,169,255,103,108,1,214,185,51 114,174,103,230,180,233,215,215,100,32,55,46,124,185,89,193,103,169,239,124,17,94 239,253,145,89,17,132,150,223,96,116,94,28,5,26,55,244,26,132,51,233,46,169,132 223,239,233,36,116,226,239,253,44,77,193,28,132,44,233,255,230,185,253,190,172,85 26,51,160,174,96,172,38,1,214,255,244,36,114,214,214,160,253,235,106,235,55,32,38 94,5,108,235,100,190,150,26,51,226,180,185,103,55,255,235,28,85,3,150,150,94,145 15,169,106,3,28,77,15,244,59,180,38,145,223,167,180,233,172,114,106,226,38,15,150 169,116,116,244,124,32,215,214,108,59,106,96,215,1,167,44,174,255,160,172,100,180 28,145,167,124,114,244,3,89,160,167,156,223,15,17,230,36,3,156,77,132,5,145,89 17,103,114
X^{129}	0,1,23,145,8,15,53,181,184,166,221,68,252,228,179,44,64,85,231,117,120,107,245,97 181,191,96,237,193,205,62,181,169,166,143,7,89,80,85,219,166,182,242,101,26,12,100 245,179,168,37,185,115,110,207,85,241,245,21,150,125,127,179,54,58,115,231,41,146 221,101,173,107,61,196,21,143,223,10,221,231,186,138,80,127,36,56,228,251,185,228,33 47,107,26,217,193,134,45,237,145,208,87,145,39,127,185,102,59,101,143,86,70,21,26 206,38,115,80,130,237,161,195,8,193,139,197,8,33,252,39,125,89,130,117,41,12,206 120,61,56,252,102,37,242,59,68,10,186,117,38,110,146,68,86,7,150,70,120,47,89,138 110,58,217,12,196,150,195,15,134,205,15,197,96,45,208,23,87,23,96,161,205,139,7,223 242,173,251,37,36,125,241,100,61,47,41,186,207,219,53,191,139,134,161,45,53,62,191 62,195,197,87,208,1,1,54,168,56,33,146,10,182,169,219,64,38,58,251,102,44,54,168,44 39,36,196,70,97,100,207,64,130,138,223,86,184,182,241,97,206,217,173,59,184,169
S_A81	0,1,0,6,0,12,2,9,0,24,50,45,4,17,52,38,0,48,47,24,100,89,73,115,8,33,21,59,104,76 119,84,0,96,229,130,93,48,186,208,200,177,31,97,145,229,68,55,16,65,218,140,41,117 225,186,208,152,40,103,237,168,23,85,0,192,76,139,203,6,133,79,185,96,199,25,118,162 10,217,146,99,241,7,61,193,92,167,35,203,114,157,136,109,219,57,32,129,137,47,182,26 29,182,81,233,202,117,195,118,90,232,162,50,36,179,80,205,212,78,219,82,111,225,45 169,155,24,0,130,128,5,152,23,26,146,151,12,37,185,11,157,187,42,115,192,220,104,143 49,34,155,236,70,113,220,20,179,139,43,38,197,67,167,227,13,132,109,121,131,46,211 184,79,237,29,69,151,15,218,228,59,172,116,18,217,106,166,183,113,205,12,64,3,140 200,19,93,221,148,110,52,144,205,57,110,197,149,161,211,66,55,150,233,119,15,135,236 86,58,180,210,103,6,70,100,111,74,72,103,99,75,160,155,187,135,170,156,179,130,183 164,177,165,221,195,217,192,89,83,109,96,55,48,1,1
S_81	0,6,57,26,104,184,82,167,164,40,240,89,206,148,153,230,19,70,15,127,0,131,31,185,188 99,205,55,173,164,223,243,78,189,120,174,3,38,54,54,17,104,74,22,94,241,6,140,53 149,38,163,3,117,19,64,97,75,31,16,85,169,40,241,147,252,152,210,18,171,26,134,145 116,247,55,18,33,119,97,74,118,100,125,176,90,157,82,67,245,0,147,187,219,251,190 94,196,90,229,250,182,253,148,167,183,206,251,1,199,107,136,239,38,206,34,48,47,18 40,29,94,81,55,192,85,143,63,69,192,199,103,118,37,247,129,193,206,46,4,240,41,28 224,218,12,125,142,146,146,54,19,85,9,159,230,31,149,214,121,105,25,228,177,127,217 241,114,22,236,246,41,2,46,225,232,158,189,54,48,243,6,88,136,234,67,47,163,133,250 67,25,102,138,214,31,188,134,15,16,68,34,153,218,156,44,66,215,51,140,166,60,146 251,4,72,26,47,226,242,185,90,66,132,201,208,118,74,54,249,138,96,16,131,194,116 237,168,60,92,244,190,110,1,112,236,233,80,38,230,209,52,160,182,84,103

S-box	LUT
S_82	0,0,0,132,0,175,87,124,0,117,131,114,194,24,22,72,0,63,5,190,80,192,2,22,209,155,87 153,67,166,146,243,0,61,70,255,49,163,32,54,54,126,243,63,197,34,87,52,84,86,23,145 53,152,33,8,179,196,115,128,16,200,135,219,0,156,6,30,243,192,162,21,145,120,20,121 160,230,114,176,184,27,187,156,27,23,79,199,248,46,120,42,153,224,78,179,74,235,10 47,136,134,159,21,237,57,46,126,237,150,121,134,166,56,227,249,52,5,38,147,208,59,22 121,128,196,17,209,0,159,85,78,30,46,28,168,231,13,49,95,59,126,186,123,217,121,137 173,151,152,144,27,239,58,60,109,99,25,231,25,234,72,249,223,197,200,129,8,59,236 171,248,214,174,17,237,103,250,113,104,24,42,89,239,103,143,242,158,218,157,24,219 239,236,188,59,2,174,6,46,153,239,73,187,182,111,49,108,142,178,216,96,51,160,50,37 41,96,252,49,86,176,212,182,79,113,90,224,147,2,209,196,15,68,153,86,17,245,208,176 122,123,106,239,246,88,177,155,235,159,120,136,165,126,97,62
S_83	0,0,0,15,0,71,50,122,0,118,239,150,165,148,120,70,0,93,25,75,171,177,128,149,104,67 158,186,102,10,162,193,0,53,115,73,82,32,19,110,117,54,233,165,130,134,44,39,169,193 195,164,80,127,8,40,180,170,49,32,232,177,95,9,0,191,226,82,96,152,176,71,25,208,20 210,220,82,227,98,147,113,104,133,88,253,145,59,226,118,246,109,140,95,170,118,130,8 19,150,176,125,19,209,238,18,144,99,121,194,53,129,184,111,48,232,33,177,155,4,188 29,219,117,128,102,213,60,0,190,192,113,27,226,233,31,201,1,230,33,119,248,106,234 176,83,105,133,0,164,235,64,17,132,39,189,4,214,0,221,8,131,187,63,65,141,192,3,180 73,232,26,88,226,54,131,17,199,187,98,243,98,107,245,197,101,128,47,130,101,245,29 176,177,146,156,203,141,219,146,96,23,173,213,190,142,65,126,147,207,168,251,67,88 74,94,43,1,255,218,94,51,184,218,58,14,107,80,19,96,112,12,159,221,33,108,19,22,159 149,176,217,248,158,50,28,72,105,125,98,218,202,90,2,207,152
X^3 truncated	0,1,8,15,64,85,120,107,17,88,185,246,209,140,73,18,136,153,209,198,234,239,131,128,255 166,6,89,29,80,212,159,98,97,174,171,187,172,71,86,39,108,75,6,126,33,34,123,203,216 86,67,48,55,157,156,232,179,213,136,147,220,158,215,1,200,25,214,82,143,122,161,250,123 66,197,41,188,161,50,124,165,53,234,13,192,116,191,225,112,8,159,16,149,201,74,47,228 243,62,229,58,9,208,128,3,252,121,202,93,134,23,115,168,254,35,155,84,38,239,186,41,151 2,210,85,207,78,76,252,55,129,140,40,199,101,197,61,30,224,133,105,110,132,167,7,141,43 69,241,95,237,72,160,194,44,42,214,144,106,181,7,10,190,236,74,99,195,104,146,119,139 177,95,158,118,127,221,145,53,4,178,218,106,196,46,138,102,63,193,65,185,120,0,19,109 171,199,240,154,27,43,208,230,72,108,179,145,102,14,92,50,151,235,157,231,99,67,249 223,18,38,184,138,205,183,98,30,135,233,24,112,250,200,245,193,48,22,15,47,242,152,12 96,154,228,84,44,163,129,253,217,75,125,37,21

9-bit S-box implementations

χ_9

A= 9 X= 18

$l_0 = x_8 \oplus x_6;$
$l_1 = x_7 \oplus x_5;$
$l_2 = x_6 \oplus x_4;$
$l_3 = x_5 \oplus x_3;$
$l_4 = x_4 \oplus x_2;$
$l_5 = x_8 \oplus x_1;$
$l_6 = x_3 \oplus x_1;$
$l_7 = x_7 \oplus x_0;$
$l_8 = x_2 \oplus x_0;$
$q_0 = (x_2 \times x_1);$
$y_0 = q_0 \oplus l_8;$
$q_1 = (x_8 \times x_7);$
$y_6 = q_1 \oplus l_0;$
$q_2 = (x_7 \times x_6);$
$y_5 = q_2 \oplus l_1;$
$q_3 = (x_5 \times x_4);$
$y_3 = q_3 \oplus l_3;$
$q_4 = (x_3 \times x_2);$
$y_1 = q_4 \oplus l_6;$
$q_5 = (x_6 \times x_5);$
$y_4 = q_5 \oplus l_2;$
$q_6 = (x_4 \times x_3);$
$y_2 = q_6 \oplus l_4;$
$q_7 = (x_8 \times x_0);$
$y_7 = q_7 \oplus l_7;$
$q_8 = (x_1 \times x_0);$
$y_8 = q_8 \oplus l_5;$

X^3

A=19 X=94

$l_0 = x_7 \oplus x_6;$
$l_1 = x_8 \oplus x_5;$
$l_2 = x_7 \oplus x_5;$
$l_3 = x_8 \oplus x_3;$
$l_4 = x_4 \oplus x_3;$
$l_5 = x_4 \oplus x_2;$
$l_6 = x_5 \oplus x_1;$
$l_7 = x_2 \oplus x_1;$
$l_8 = l_1 \oplus l_4;$
$l_9 = x_6 \oplus l_4;$
$l_{10} = x_8 \oplus l_2;$
$l_{11} = l_{10} \oplus x_2;$
$l_{12} = l_3 \oplus l_9;$
$l_{13} = l_{12} \oplus l_6;$
$l_{14} = l_9 \oplus x_1;$
$l_{15} = l_{13} \oplus x_0;$
$l_{16} = l_3 \oplus x_2;$
$l_{17} = l_{14} \oplus x_0;$
$l_{18} = l_5 \oplus l_6;$
$l_{19} = l_{18} \oplus l_{17};$
$l_{20} = x_8 \oplus l_5;$
$l_{21} = l_2 \oplus l_3;$
$l_{22} = l_{20} \oplus l_{17};$
$l_{23} = l_{21} \oplus l_{19};$
$l_{24} = l_{10} \oplus l_7;$
$l_{25} = l_7 \oplus l_{22};$
$l_{26} = x_5 \oplus l_{11};$
$l_{27} = x_7 \oplus l_{23};$
$l_{28} = x_6 \oplus l_{17};$
$l_{29} = x_8 \oplus l_{11};$
$l_{30} = l_{29} \oplus l_{18};$
$l_{31} = x_3 \oplus l_5;$
$l_{32} = l_1 \oplus x_4;$
$l_{33} = l_{24} \oplus l_{23};$
$l_{34} = l_{12} \oplus l_{30};$
$l_{35} = l_{20} \oplus l_{23};$
$l_{36} = l_{12} \oplus l_{35};$
$l_{37} = l_0 \oplus l_{29};$
$l_{38} = l_4 \oplus l_{18};$

$$
\begin{aligned}
q_0 &= (x_8 \times l_0);\\
q_1 &= (l_{37} \times l_{20}) \oplus x_2;\\
q_2 &= (x_4 \times l_7);\\
l_{39} &= q_0 \oplus q_2;\\
q_3 &= (x_7 \times l_{15});\\
l_{40} &= q_1 \oplus q_3;\\
y_0 &= l_{39} \oplus l_{40} \oplus l_{25};\\
q_4 &= (l_2 \times l_4);\\
q_5 &= (x_6 \times x_1);\\
q_6 &= (l_5 \times l_{33});\\
l_{41} &= q_4 \oplus q_6;\\
y_4 &= q_0 \oplus l_{41} \oplus q_3 \oplus q_5 \oplus l_3;\\
q_7 &= (l_{10} \times l_9);\\
q_8 &= (l_{26} \times l_{28});\\
y_8 &= l_{40} \oplus q_2 \oplus q_7 \oplus q_8 \oplus q_6 \oplus x_6;\\
q_9 &= (l_{29} \times l_8) \oplus x_5;\\
q_{10} &= (l_{16} \times l_{14}) \oplus x_3;\\
y_5 &= q_0 \oplus l_{40} \oplus q_9 \oplus q_{10} \oplus q_4 \oplus l_1;\\
q_{11} &= (l_{21} \times l_{13}) \oplus l_1;\\
q_{12} &= (l_{24} \times l_{17}) \oplus x_1;\\
y_1 &= l_{39} \oplus q_5 \oplus q_{11} \oplus q_8 \oplus q_{12} \oplus x_5;\\
q_{13} &= (l_{32} \times l_{22}) \oplus x_8;\\
q_{14} &= (l_6 \times l_{27});\\
y_2 &= l_{41} \oplus q_1 \oplus q_5 \oplus q_{13} \oplus q_7 \oplus q_{14} \oplus x_7;\\
q_{15} &= (l_{31} \times l_{36});\\
q_{16} &= (l_{34} \times l_{19}) \oplus x_6;\\
y_6 &= l_{39} \oplus q_9 \oplus q_{13} \oplus q_{15} \oplus q_{16} \oplus q_{12} \oplus l_{11};\\
q_{17} &= (l_{38} \times l_{23}) \oplus x_2;\\
y_3 &= l_{41} \oplus q_3 \oplus q_{13} \oplus q_{10} \oplus q_{16} \oplus q_{17} \oplus l_{30};\\
q_{18} &= (x_5 \times l_{35});\\
y_7 &= q_7 \oplus q_1 \oplus q_9 \oplus q_{18} \oplus q_{16} \oplus q_{14} \oplus x_4;
\end{aligned}
$$

X^5

A= 20 X= 109

$$
\begin{aligned}
l_0 &= x_7 \oplus x_6;\\
l_1 &= x_6 \oplus x_5;\\
l_2 &= x_8 \oplus x_4;\\
l_3 &= x_6 \oplus x_3;\\
l_4 &= x_8 \oplus x_2;\\
l_5 &= x_5 \oplus x_2;\\
l_6 &= l_5 \oplus x_1;\\
l_7 &= x_7 \oplus l_2;\\
l_8 &= x_6 \oplus l_5;\\
l_9 &= l_7 \oplus l_4;\\
l_{10} &= x_8 \oplus x_3;\\
l_{11} &= l_{10} \oplus x_0;\\
l_{12} &= l_0 \oplus x_4;\\
l_{13} &= l_5 \oplus l_{11};\\
l_{14} &= l_1 \oplus l_9;\\
l_{15} &= l_9 \oplus x_1;\\
l_{16} &= l_1 \oplus x_1;\\
l_{17} &= l_{16} \oplus x_0;\\
l_{18} &= l_1 \oplus l_{10};\\
l_{19} &= l_{18} \oplus l_{15};\\
l_{20} &= x_6 \oplus l_{18};\\
l_{21} &= l_0 \oplus l_4;\\
l_{22} &= l_{21} \oplus l_{19};\\
l_{23} &= l_{21} \oplus l_{11};\\
l_{24} &= l_{20} \oplus l_{16};\\
l_{25} &= x_4 \oplus l_6;\\
l_{26} &= x_7 \oplus l_{14};\\
l_{27} &= l_3 \oplus l_{17};\\
l_{28} &= l_{10} \oplus l_{14};\\
l_{29} &= x_7 \oplus l_{20};\\
l_{30} &= l_3 \oplus l_4;\\
l_{31} &= x_8 \oplus l_{29};\\
l_{32} &= l_{31} \oplus l_{11};\\
l_{33} &= x_6 \oplus l_{31};\\
l_{34} &= x_5 \oplus l_{27};\\
l_{35} &= x_7 \oplus l_{32};\\
l_{36} &= l_{12} \oplus x_3;\\
l_{37} &= l_{26} \oplus l_{32};\\
l_{38} &= x_6 \oplus l_{37};\\
l_{39} &= x_8 \oplus l_0;\\
l_{40} &= x_8 \oplus l_{27};\\
l_{41} &= x_6 \oplus l_2;\\
l_{42} &= l_{10} \oplus l_{38};\\
l_{43} &= l_{21} \oplus l_{38};\\
l_{44} &= x_6 \oplus l_{34};\\
l_{45} &= l_2 \oplus l_{32};\\
l_{46} &= l_{21} \oplus l_{45};\\
q_0 &= (l_{35} \times l_{33}) \oplus x_5;\\
q_1 &= (l_{44} \times l_{14}) \oplus x_6;\\
q_2 &= (l_{22} \times l_{42}) \oplus x_4 \oplus x_3;\\
q_3 &= (l_{23} \times l_{40}) \oplus x_3 \oplus x_0;\\
y_0 &= q_0 \oplus q_1 \oplus q_2 \oplus q_3 \oplus x_0;\\
q_4 &= (l_3 \times l_{12}) \oplus x_6;\\
l_{47} &= q_1 \oplus q_4;\\
q_5 &= (l_1 \times l_{30}) \oplus x_6;\\
q_6 &= (l_{27} \times l_{38}) \oplus x_0;\\
y_8 &= q_4 \oplus q_5 \oplus q_2 \oplus q_6 \oplus x_7;\\
q_7 &= (l_1 \times l_4);\\
q_8 &= (l_{20} \times l_{37}) \oplus x_8;\\
l_{48} &= q_7 \oplus q_8;\\
y_5 &= q_4 \oplus l_{48} \oplus q_0 \oplus l_{19};\\
q_9 &= (l_{18} \times l_{26}) \oplus l_1;\\
q_{10} &= (l_{34} \times l_{46}) \oplus x_0;\\
l_{49} &= q_3 \oplus q_9;\\
y_1 &= l_{47} \oplus q_5 \oplus q_9 \oplus l_{48} \oplus q_{10} \oplus l_{28};\\
q_{11} &= (l_5 \times l_7);\\
q_{12} &= (l_{39} \times l_{43}) \oplus x_6;\\
y_7 &= q_7 \oplus q_{11} \oplus q_{12} \oplus q_3 \oplus q_6 \oplus l_1;\\
q_{13} &= (l_2 \times l_{29}) \oplus x_8;\\
q_{14} &= (l_{15} \times l_{32}) \oplus x_7;\\
y_2 &= q_{13} \oplus l_{47} \oplus q_8 \oplus q_{14} \oplus q_{10} \oplus l_{41};\\
q_{15} &= (l_8 \times l_9) \oplus x_2;\\
q_{16} &= (l_{24} \times l_{45});\\
y_3 &= l_{47} \oplus q_{12} \oplus q_{15} \oplus q_{16} \oplus q_{10} \oplus l_0;\\
q_{17} &= (l_{36} \times l_{17}) \oplus x_6;\\
q_{18} &= (l_6 \times l_{11});\\
l_{50} &= q_{16} \oplus q_{18};\\
y_4 &= l_{49} \oplus q_{17} \oplus l_{50} \oplus q_{13} \oplus x_8;\\
q_{19} &= (l_{25} \times l_{13}) \oplus x_5 \oplus x_2;\\
y_6 &= l_{49} \oplus q_2 \oplus l_{50} \oplus q_{14} \oplus q_{19} \oplus l_{31};
\end{aligned}
$$

X^9

A= 18 X= 100

$$
\begin{aligned}
l_0 &= x_8 \oplus x_7;\\
l_1 &= x_8 \oplus x_6;\\
l_2 &= x_7 \oplus x_6;\\
l_3 &= x_6 \oplus x_4;\\
l_4 &= x_5 \oplus x_4;\\
l_5 &= x_7 \oplus x_3;\\
l_6 &= x_6 \oplus x_3;\\
l_7 &= x_5 \oplus x_1;\\
l_8 &= x_7 \oplus x_0;\\
l_9 &= x_4 \oplus x_0;\\
l_{10} &= x_5 \oplus l_6;\\
l_{11} &= l_{10} \oplus x_0;\\
l_{12} &= l_2 \oplus l_4;\\
l_{13} &= l_6 \oplus l_7;\\
l_{14} &= l_1 \oplus x_1;\\
l_{15} &= l_2 \oplus l_{10};\\
l_{16} &= l_{15} \oplus x_2;\\
l_{17} &= x_7 \oplus l_7;\\
l_{18} &= l_6 \oplus x_2;\\
l_{19} &= x_8 \oplus l_5;\\
l_{20} &= l_{16} \oplus l_8;\\
l_{21} &= l_{16} \oplus l_{14};\\
l_{22} &= l_3 \oplus l_5;\\
l_{23} &= x_8 \oplus l_7;\\
l_{24} &= l_{23} \oplus l_{20};\\
l_{25} &= l_{12} \oplus l_{19};\\
l_{26} &= l_{22} \oplus l_{20};\\
l_{27} &= l_{25} \oplus x_0;\\
l_{28} &= l_{15} \oplus l_{21};\\
l_{29} &= l_{13} \oplus l_{26};\\
l_{30} &= l_{16} \oplus l_{17};\\
l_{31} &= x_8 \oplus l_6;\\
l_{32} &= x_7 \oplus l_{22};\\
l_{33} &= l_{23} \oplus l_{28};\\
l_{34} &= l_6 \oplus l_8;\\
l_{35} &= l_2 \oplus l_{26};\\
l_{36} &= l_{19} \oplus l_{18};\\
l_{37} &= x_8 \oplus l_{15};\\
l_{38} &= x_5 \oplus l_{29};\\
l_{39} &= x_5 \oplus l_{21};\\
l_{40} &= l_{19} \oplus l_{10};\\
l_{41} &= l_{15} \oplus l_{27};\\
l_{42} &= l_0 \oplus x_5;\\
l_{43} &= l_5 \oplus l_{30};\\
q_0 &= (l_1 \times l_{37}) \oplus x_8;\\
q_1 &= (x_8 \times l_{16});\\
q_2 &= (l_{22} \times l_{20}) \oplus x_3;\\
l_{44} &= q_1 \oplus q_2;\\
y_0 &= q_0 \oplus l_{44} \oplus l_{38};\\
q_3 &= (l_2 \times l_{42}) \oplus x_7;\\
q_4 &= (l_6 \times l_{33}) \oplus x_6;\\
q_5 &= (l_{14} \times l_{26}) \oplus x_6;\\
l_{45} &= q_4 \oplus q_5;\\
y_8 &= q_3 \oplus l_{45} \oplus l_{18};\\
q_6 &= (l_3 \times l_{31}) \oplus x_6;\\
q_7 &= (l_{11} \times l_{39}) \oplus l_6;\\
y_7 &= q_6 \oplus l_{44} \oplus l_{45} \oplus q_7 \oplus l_{25};\\
q_8 &= (l_0 \times l_{32});\\
q_9 &= (l_8 \times l_{21}) \oplus x_7;\\
l_{46} &= q_0 \oplus q_9;\\
l_{47} &= q_6 \oplus q_8;\\
y_2 &= q_8 \oplus q_7 \oplus q_9 \oplus l_3;\\
q_{10} &= (l_5 \times l_{43}) \oplus x_7;\\
q_{11} &= (l_7 \times l_{27}) \oplus x_5;\\
y_6 &= l_{46} \oplus l_{47} \oplus q_{11} \oplus q_{10} \oplus l_4;\\
q_{12} &= (l_{17} \times l_{24}) \oplus x_1;\\
q_{13} &= (l_{30} \times l_{35}) \oplus x_2;\\
l_{48} &= q_{12} \oplus q_{13};\\
y_5 &= l_{46} \oplus q_5 \oplus l_{48} \oplus q_2 \oplus l_{40};\\
q_{14} &= (l_{14} \times l_{29}) \oplus x_1;\\
y_4 &= l_{47} \oplus q_2 \oplus q_{14} \oplus q_{12} \oplus l_{13};\\
q_{15} &= (l_{36} \times l_{34}) \oplus l_2;\\
q_{16} &= (l_{28} \times l_{41}) \oplus l_1;\\
l_{49} &= q_5 \oplus q_{16};\\
y_1 &= q_3 \oplus q_0 \oplus q_{15} \oplus l_{49} \oplus q_{11} \oplus x_5;\\
q_{17} &= (l_9 \times l_{12}) \oplus x_4;\\
y_3 &= q_1 \oplus q_{10} \oplus l_{49} \oplus l_{48} \oplus q_{17} \oplus l_{19};
\end{aligned}
$$

X^{65}

A= 18 X= 97

$$
\begin{aligned}
l_0 &= x_8 \oplus x_6;\\
l_1 &= x_7 \oplus x_6;\\
l_2 &= x_5 \oplus x_4;\\
l_3 &= x_7 \oplus x_3;\\
l_4 &= x_3 \oplus x_1;\\
l_5 &= x_8 \oplus l_4;\\
l_6 &= l_1 \oplus x_5;\\
l_7 &= l_6 \oplus l_5;\\
l_8 &= x_5 \oplus x_1;\\
l_9 &= l_8 \oplus x_0;\\
l_{10} &= l_4 \oplus x_0;\\
l_{11} &= x_2 \oplus l_9;\\
l_{12} &= l_1 \oplus x_2;\\
l_{13} &= l_2 \oplus l_{12};\\
l_{14} &= l_0 \oplus l_7;\\
l_{15} &= x_7 \oplus l_{13};\\
l_{48} &= x_3 \oplus x_4;\\
l_{49} &= x_0 \oplus x_6;\\
l_{16} &= l_{48} \oplus l_{49};\\
l_{17} &= l_4 \oplus l_{11};\\
l_{18} &= l_7 \oplus l_{17};\\
l_{19} &= x_6 \oplus l_3;\\
l_{20} &= x_3 \oplus l_7;\\
l_{21} &= x_8 \oplus l_{14};\\
l_{22} &= x_8 \oplus l_{19};\\
l_{23} &= x_7 \oplus x_4;\\
l_{24} &= l_{23} \oplus l_8;\\
l_{25} &= l_{19} \oplus l_{15};\\
l_{26} &= x_4 \oplus l_{19};\\
l_{27} &= l_0 \oplus l_6;\\
l_{28} &= x_7 \oplus l_8;\\
l_{29} &= x_4 \oplus l_{25};\\
l_{30} &= x_4 \oplus l_{16};\\
l_{31} &= x_7 \oplus l_2;\\
l_{32} &= x_2 \oplus l_{16};\\
l_{33} &= l_0 \oplus l_{26};\\
l_{34} &= l_{33} \oplus l_{17};\\
l_{35} &= l_{21} \oplus l_9;\\
l_{36} &= l_{33} \oplus l_{13};\\
l_{37} &= l_{13} \oplus l_{24};\\
l_{38} &= l_0 \oplus l_{15};\\
l_{39} &= l_{26} \oplus l_8;\\
l_{40} &= x_8 \oplus l_{15};\\
l_{41} &= x_6 \oplus l_{34};\\
l_{42} &= x_5 \oplus l_{37};\\
q_0 &= (l_{31} \times l_9) \oplus x_5;\\
q_1 &= (l_{10} \times l_{13});\\
q_2 &= (l_{16} \times l_{25}) \oplus l_{48};\\
y_0 &= q_0 \oplus q_1 \oplus q_2 \oplus l_{41};\\
q_3 &= (l_2 \times l_6) \oplus x_5;\\
q_4 &= (l_{22} \times l_{40}) \oplus l_0;\\
q_5 &= (l_{28} \times l_{32});\\
l_{43} &= q_4 \oplus q_5;\\
y_6 &= q_3 \oplus l_{43} \oplus l_3;\\
q_6 &= (l_1 \times l_5);\\
q_7 &= (l_{29} \times l_{35}) \oplus l_3;\\
l_{44} &= q_3 \oplus q_6;\\
l_{45} &= q_0 \oplus l_{44};\\
y_4 &= l_{43} \oplus q_1 \oplus q_7 \oplus q_6 \oplus x_8;\\
q_8 &= (l_{27} \times l_{15}) \oplus x_5;\\
q_9 &= (l_{33} \times l_{42});\\
q_{10} &= (l_{26} \times l_{18}) \oplus l_1;\\
l_{46} &= q_8 \oplus q_9;\\
l_{47} &= l_{46} \oplus q_{10};\\
y_2 &= l_{47} \oplus l_{39};\\
q_{11} &= (l_{37} \times l_{21}) \oplus x_1;\\
q_{12} &= (l_{11} \times l_{36}) \oplus x_5 \oplus x_2;\\
y_8 &= l_{44} \oplus q_{11} \oplus q_{12} \oplus x_3;\\
q_{13} &= (l_{38} \times l_7) \oplus x_8 \oplus x_5;\\
q_{14} &= (l_{15} \times l_{24}) \oplus l_2;\\
y_7 &= l_{45} \oplus q_{13} \oplus q_{14} \oplus q_{10} \oplus q_2 \oplus l_4;\\
q_{15} &= (x_2 \times l_0);\\
y_3 &= q_{15} \oplus q_{14} \oplus q_{11} \oplus q_7 \oplus l_{19};\\
q_{16} &= (l_{30} \times l_{34}) \oplus x_0;\\
y_5 &= l_{47} \oplus q_0 \oplus q_{13} \oplus q_{12} \oplus q_{16} \oplus l_{14};\\
q_{17} &= (l_{26} \times l_{20}) \oplus l_1;\\
y_1 &= l_{45} \oplus q_{17} \oplus q_8 \oplus q_7 \oplus q_{16} \oplus l_{12};
\end{aligned}
$$

$X^3 \oplus X^5$

A= 18 X= 100

$$
\begin{aligned}
l_0 &= x_7 \oplus x_6;\\
l_1 &= x_6 \oplus x_4;\\
l_2 &= x_7 \oplus x_2;\\
l_3 &= x_6 \oplus x_2;\\
l_4 &= x_2 \oplus x_0;\\
l_5 &= x_5 \oplus x_3;\\
l_6 &= l_5 \oplus x_2;\\
l_7 &= x_8 \oplus x_7;\\
l_8 &= l_7 \oplus x_3;\\
l_9 &= x_8 \oplus l_5;\\
l_{10} &= x_5 \oplus x_1;\\
l_{11} &= l_{10} \oplus x_0;\\
l_{12} &= l_7 \oplus x_5;\\
l_{13} &= l_{12} \oplus l_4;\\
l_{14} &= x_4 \oplus l_8;\\
l_{15} &= l_{14} \oplus l_2;\\
l_{16} &= l_0 \oplus l_{15};\\
l_{17} &= l_1 \oplus l_9;\\
l_{18} &= l_9 \oplus l_{13};\\
l_{19} &= l_7 \oplus l_1;\\
l_{20} &= l_7 \oplus x_1;\\
l_{21} &= x_4 \oplus l_{15};\\
l_{22} &= l_{17} \oplus l_{10};\\
l_{23} &= l_0 \oplus l_5;\\
l_{24} &= l_6 \oplus l_{15};\\
l_{25} &= l_1 \oplus x_0;\\
l_{26} &= l_{12} \oplus l_{17};\\
l_{27} &= l_7 \oplus l_{11};\\
l_{28} &= l_{14} \oplus l_{18};\\
l_{29} &= l_9 \oplus l_{16};\\
l_{30} &= l_6 \oplus l_{18};\\
l_{31} &= l_6 \oplus l_{10};\\
l_{32} &= l_{26} \oplus l_{27};\\
l_{33} &= l_{13} \oplus l_{32};\\
l_{34} &= l_{24} \oplus x_1;\\
l_{35} &= l_0 \oplus l_8;\\
l_{36} &= l_{26} \oplus x_0;\\
l_{37} &= l_1 \oplus x_1;\\
l_{38} &= l_{14} \oplus l_{27};\\
l_{39} &= x_7 \oplus l_6;\\
l_{40} &= l_{23} \oplus l_{33};\\
l_{41} &= x_5 \oplus l_1;\\
l_{42} &= l_{35} \oplus x_0;\\
q_0 &= (l_{23} \times l_{15}) \oplus x_3;\\
q_1 &= (l_{31} \times l_{17}) \oplus x_3;\\
q_2 &= (l_{35} \times l_{27}) \oplus x_8;\\
y_3 &= q_0 \oplus q_1 \oplus q_2 \oplus l_{37};\\
q_3 &= (l_1 \times l_8);\\
q_4 &= (x_5 \times l_3);\\
q_5 &= (x_0 \times l_{34});\\
l_{43} &= q_0 \oplus q_4;\\
y_2 &= q_3 \oplus l_{43} \oplus q_5 \oplus l_{19};\\
q_6 &= (l_{24} \times l_{26}) \oplus x_4;\\
q_7 &= (l_6 \times l_{36}) \oplus x_3;\\
q_8 &= (l_4 \times l_{39}) \oplus x_2;\\
l_{44} &= q_4 \oplus q_7;\\
l_{45} &= l_{44} \oplus q_8;\\
y_1 &= q_6 \oplus l_{45} \oplus l_{16};\\
q_9 &= (l_9 \times l_{36}) \oplus x_3;\\
q_{10} &= (l_{25} \times l_{18}) \oplus x_0;\\
l_{46} &= q_1 \oplus q_{10};\\
l_{47} &= q_6 \oplus q_9;\\
y_4 &= l_{46} \oplus q_9 \oplus q_5 \oplus x_3;\\
q_{11} &= (l_{30} \times l_{42}) \oplus x_0;\\
q_{12} &= (l_{11} \times l_{29}) \oplus x_5;\\
l_{48} &= q_{11} \oplus q_{12};\\
y_5 &= l_{48} \oplus l_{46} \oplus l_{45} \oplus l_{33};\\
q_{13} &= (l_{25} \times l_{23}) \oplus x_6;\\
q_{14} &= (l_2 \times l_{32});\\
l_{49} &= q_{13} \oplus q_{14};\\
y_7 &= l_{47} \oplus ql49 \oplus l_{48} \oplus l_{41};\\
q_{15} &= (l_{13} \times l_{38}) \oplus x_5 \oplus x_0;\\
y_6 &= l_{47} \oplus q_{14} \oplus q_{12} \oplus q_8 \oplus q_5 \oplus q_{15} \oplus l_{21};\\
q_{16} &= (l_{20} \times l_{40}) \oplus x_1;\\
y_8 &= l_{43} \oplus q_{16} \oplus q_2 \oplus q_9 \oplus q_{12} \oplus l_0;\\
q_{17} &= (l_{28} \times l_{22}) \oplus x_8 \oplus x_4;\\
y_0 &= q_3 \oplus q_4 \oplus q_{11} \oplus l_{49} \oplus q_{17} \oplus q_{15} \oplus l_{35};
\end{aligned}
$$

$X^3 \oplus X^9$

A= 18 X= 96

$$
\begin{aligned}
l_0 &= x_7 \oplus x_6;\\
l_1 &= x_6 \oplus x_5;\\
l_2 &= x_8 \oplus x_3;\\
l_3 &= x_6 \oplus x_3;\\
l_4 &= x_3 \oplus x_2;\\
l_5 &= x_2 \oplus x_1;\\
l_6 &= x_7 \oplus x_0;\\
l_7 &= x_8 \oplus l_6;\\
l_8 &= x_5 \oplus x_2;\\
l_9 &= l_8 \oplus l_7;\\
l_{10} &= l_0 \oplus x_4;\\
l_{11} &= l_{10} \oplus x_1;\\
l_{12} &= x_3 \oplus l_{11};\\
l_{13} &= x_8 \oplus l_0;\\
l_{14} &= l_{13} \oplus l_8;\\
l_{15} &= x_4 \oplus l_8;\\
l_{16} &= l_1 \oplus l_{10};\\
l_{17} &= l_{16} \oplus l_2;\\
l_{18} &= x_5 \oplus l_6;\\
l_{19} &= l_{15} \oplus l_4;\\
l_{20} &= l_{15} \oplus l_{12};\\
l_{21} &= l_0 \oplus l_{11};\\
l_{22} &= l_{21} \oplus l_7;\\
l_{23} &= l_2 \oplus x_0;\\
l_{24} &= l_1 \oplus l_2;\\
l_{25} &= l_{17} \oplus l_6;\\
l_{26} &= l_{13} \oplus l_{23};\\
l_{27} &= l_{13} \oplus x_1;\\
l_{28} &= l_{13} \oplus l_{12};\\
l_{29} &= l_4 \oplus l_{23};\\
l_{30} &= x_7 \oplus l_8;\\
l_{31} &= l_{12} \oplus l_7;\\
l_{32} &= l_{15} \oplus l_6;\\
l_{33} &= l_1 \oplus l_{20};\\
l_{34} &= l_2 \oplus l_{15};\\
l_{35} &= l_{14} \oplus l_{21};\\
l_{36} &= x_8 \oplus l_{27};\\
l_{37} &= l_1 \oplus l_{19};\\
l_{38} &= l_{19} \oplus l_{18};\\
l_{39} &= l_{28} \oplus l_{33};\\
l_{40} &= l_{17} \oplus l_{12};\\
l_{41} &= l_{20} \oplus l_{26};\\
l_{42} &= l_{29} \oplus l_{22};\\
l_{43} &= l_{38} \oplus l_{41};\\
q_0 &= (l_{30} \times l_{27}) \oplus x_7;\\
q_1 &= (l_3 \times l_7);\\
q_2 &= (l_{38} \times l_{14}) \oplus x_7;\\
l_{44} &= q_1 \oplus q_2;\\
y_4 &= q_0 \oplus l_{44} \oplus l_{40};\\
q_3 &= (l_2 \times l_{13}) \oplus x_8;\\
q_4 &= (l_1 \times l_{12}) \oplus x_6;\\
q_5 &= (l_{23} \times l_{32}) \oplus x_0;\\
l_{45} &= q_4 \oplus q_5;\\
y_5 &= q_3 \oplus q_2 \oplus l_{45} \oplus l_0;\\
q_6 &= (l_{16} \times l_{37}) \oplus x_4;\\
q_7 &= (l_{36} \times l_9) \oplus x_7;\\
l_{46} &= q_6 \oplus q_7;\\
l_{47} &= l_{45} \oplus l_{46};\\
q_8 &= (l_{18} \times l_{31}) \oplus x_0;\\
l_{50} &= q_0 \oplus q_8;\\
y_6 &= l_{47} \oplus l_{50} \oplus l_{39};\\
q_9 &= (l_{41} \times l_{34}) \oplus l_8;\\
q_{10} &= (l_{11} \times l_{26}) \oplus l_0;\\
l_{48} &= q_5 \oplus q_9;\\
l_{49} &= l_{48} \oplus q_{10};\\
y_8 &= l_{49} \oplus l_4;\\
q_{11} &= (x_8 \times l_{20});\\
q_{12} &= (l_5 \times l_{15}) \oplus x_2;\\
y_3 &= q_{11} \oplus q_{12} \oplus l_{44} \oplus q_5 \oplus l_{28};\\
q_{13} &= (l_{33} \times l_{25}) \oplus x_3;\\
y_2 &= l_{50} \oplus l_{49} \oplus q_3 \oplus q_{13} \oplus l_{10};\\
q_{14} &= (l_{42} \times l_{17}) \oplus x_7 \oplus x_4;\\
q_{15} &= (l_{19} \times l_{22}) \oplus x_4;\\
y_7 &= q_3 \oplus q_{14} \oplus q_{12} \oplus q_{15} \oplus l_{44} \oplus l_{24};\\
q_{16} &= (x_2 \times l_6);\\
y_0 &= q_6 \oplus q_{11} \oplus q_4 \oplus q_{16} \oplus q_{10} \oplus q_{13} \oplus l_{35};\\
q_{17} &= (l_{29} \times l_{43}) \oplus x_2;\\
y_1 &= l_{47} \oplus q_0 \oplus q_{14} \oplus q_{17};
\end{aligned}
$$

$X^3 \oplus X^5 \oplus X^{17}$

A = 18 X = 93

$$
\begin{aligned}
l_0 &= x_7 \oplus x_6;\\
l_1 &= x_7 \oplus x_5;\\
l_2 &= x_6 \oplus x_4;\\
l_3 &= x_6 \oplus x_2;\\
l_4 &= x_4 \oplus x_2;\\
l_5 &= x_3 \oplus x_2;\\
l_6 &= x_4 \oplus x_1;\\
l_7 &= l_1 \oplus x_1;\\
l_8 &= x_5 \oplus l_2;\\
l_9 &= l_8 \oplus x_3;\\
l_{10} &= l_3 \oplus x_0;\\
l_{11} &= x_5 \oplus l_9;\\
l_{12} &= x_8 \oplus x_7;\\
l_{13} &= l_6 \oplus x_0;\\
l_{14} &= l_{12} \oplus l_{13};\\
l_{15} &= x_4 \oplus l_9;\\
l_{16} &= l_1 \oplus l_5;\\
l_{17} &= x_2 \oplus l_{13};\\
l_{18} &= l_2 \oplus l_{16};\\
l_{19} &= l_0 \oplus l_{11};\\
l_{20} &= l_{19} \oplus x_0;\\
l_{21} &= l_3 \oplus x_1;\\
l_{22} &= x_4 \oplus l_5;\\
l_{23} &= l_{12} \oplus l_{10};\\
l_{24} &= x_1 \oplus l_{14};\\
l_{25} &= x_6 \oplus l_{16};\\
l_{26} &= l_{20} \oplus l_{23};\\
l_{27} &= x_6 \oplus l_{23};\\
l_{28} &= x_7 \oplus l_{26};\\
l_{29} &= l_{18} \oplus l_{24};\\
l_{30} &= x_8 \oplus l_{21};\\
l_{31} &= l_{19} \oplus l_7;\\
l_{32} &= l_{19} \oplus l_6;\\
l_{33} &= l_2 \oplus l_{30};\\
l_{34} &= l_2 \oplus l_{10};\\
l_{35} &= l_7 \oplus l_{30};\\
l_{36} &= l_9 \oplus l_{33};\\
l_{37} &= l_1 \oplus l_6;\\
l_{38} &= x_7 \oplus l_4;\\
l_{39} &= l_{22} \oplus l_{13};\\
l_{40} &= x_7 \oplus l_{30};\\
l_{41} &= l_0 \oplus l_{35};\\
l_{42} &= x_5 \oplus l_{27};\\
l_{43} &= l_{12} \oplus x_4;\\
q_0 &= (l_{35} \times l_{14}) \oplus l_{12};\\
q_1 &= (l_{21} \times l_{27}) \oplus x_2;\\
q_2 &= (l_{24} \times l_{36}) \oplus x_8;\\
l_{44} &= q_0 \oplus q_2;\\
l_{45} &= l_{44} \oplus q_1;\\
y_3 &= l_{45} \oplus l_{11};\\
q_3 &= (l_1 \times l_5);\\
q_4 &= (l_0 \times l_7) \oplus x_7;\\
q_5 &= (l_8 \times l_{39});\\
l_{46} &= q_4 \oplus q_5;\\
y_8 &= q_3 \oplus l_{46} \oplus l_{45} \oplus l_{31};\\
q_6 &= (l_{15} \times l_{38});\\
q_7 &= (l_{20} \times l_{40}) \oplus x_7;\\
y_5 &= l_{44} \oplus q_7 \oplus q_6 \oplus l_{37};\\
q_8 &= (l_2 \times l_{25}) \oplus x_6;\\
q_9 &= (l_{32} \times l_{23}) \oplus x_7;\\
y_6 &= q_3 \oplus q_8 \oplus q_9 \oplus q_1 \oplus q_7 \oplus l_{16};\\
q_{10} &= (l_{30} \times l_{18}) \oplus l_3;\\
y_0 &= q_{10} \oplus l_{44} \oplus q_7 \oplus l_{29};\\
q_{11} &= (x_5 \times l_{19});\\
q_{12} &= (l_{43} \times l_{34}) \oplus x_4;\\
l_{47} &= q_{10} \oplus q_{12};\\
y_4 &= q_{11} \oplus l_{47} \oplus q_5 \oplus q_7 \oplus l_3;\\
q_{13} &= (x_7 \times l_4);\\
q_{14} &= (l_{10} \times l_9) \oplus x_6;\\
y_7 &= q_{13} \oplus q_3 \oplus q_{14} \oplus q_0 \oplus q_9 \oplus l_{33};\\
q_{15} &= (l_{17} \times l_{28}) \oplus l_4;\\
y_1 &= l_{46} \oplus l_{47} \oplus q_{13} \oplus q_{15} \oplus q_1 \oplus l_{41};\\
q_{16} &= (x_6 \times l_{22});\\
q_{17} &= (l_6 \times l_{42});\\
y_2 &= q_{16} \oplus l_{47} \oplus q_{14} \oplus q_{17} \oplus q_2 \oplus l_{26};
\end{aligned}
$$

S-box	LUT
χ_9	0,129,258,131,5,132,262,135,10,139,264,137,13,140,270,143,20,149,278,151,17,144,274 147,26,155,280,153,29,156,286,159,40,169,298,171,45,172,302,175,34,163,288,161,37,164 294,167,52,181,310,183,49,176,306,179,58,187,312,185,61,188,318,191,80,209,338,211,85 212,342,215,90,219,344,217,93,220,350,223,68,197,326,199,65,192,322,195,74,203,328,201 77,204,334,207,104,233,362,235,109,236,366,239,98,227,352,225,101,228,358,231,116,245 374,247,113,240,370,243,122,251,376,249,125,252,382,255,160,33,418,35,165,36,422,39 170,43,424,41,173,44,430,47,180,53,438,55,177,48,434,51,186,59,440,57,189,60,446,63 136,9,394,11,141,12,398,15,130,3,384,1,133,4,390,7,148,21,406,23,145,16,402,19,154,27 408,25,157,28,414,31,208,81,466,83,213,84,470,87,218,91,472,89,221,92,478,95,196,69 454,71,193,64,450,67,202,75,456,73,205,76,462,79,232,105,490,107,237,108,494,111,226 99,480,97,229,100,486,103,244,117,502,119,241,112,498,115,250,123,504,121,253,124,510 127,320,321,66,323,325,324,70,327,330,331,72,329,333,332,78,335,340,341,86,343,337 336,82,339,346,347,88,345,349,348,94,351,360,361,106,363,365,364,110,367,354,355,96 353,357,356,102,359,372,373,118,375,369,368,114,371,378,379,120,377,381,380,126,383 272,273,18,275,277,276,22,279,282,283,24,281,285,284,30,287,260,261,6,263,257,256,2 259,266,267,8,265,269,268,14,271,296,297,42,299,301,300,46,303,290,291,32,289,293,292 38,295,308,309,54,311,305,304,50,307,314,315,56,313,317,316,62,319,416,417,162,419,421 420,166,423,426,427,168,425,429,428,174,431,436,437,182,439,433,432,178,435,442,443 184,441,445,444,190,447,392,393,138,395,397,396,142,399,386,387,128,385,389,388,134 391,404,405,150,407,401,400,146,403,410,411,152,409,413,412,158,415,464,465,210,467 469,468,214,471,474,475,216,473,477,476,222,479,452,453,198,455,449,448,194,451,458 459,200,457,461,460,206,463,488,489,234,491,493,492,238,495,482,483,224,481,485,484 230,487,500,501,246,503,497,496,242,499,506,507,248,505,509,508,254,511
X^3	0,1,8,15,64,85,120,107,17,88,185,246,465,396,329,274,136,409,209,454,490,239,387,128 255,422,6,345,29,336,212,415,98,97,174,171,187,172,71,86,295,364,331,262,126,33,34,123 459,216,342,67,48,311,157,412,232,435,213,392,147,476,158,471,257,456,281,470,338,399 378,417,506,379,322,453,41,188,161,50,380,165,309,234,13,448,116,447,481,112,264,159 272,149,457,74,47,228,243,62,229,58,9,208,384,259,508,377,202,93,134,23,371,168,510 35,155,340,38,495,186,297,151,258,210,341,207,334,76,252,55,129,140,40,199,101,197,61 30,224,389,361,366,388,423,7,397,43,69,497,95,493,328,160,450,44,298,214,400,106,437 263,266,446,492,330,355,451,104,146,119,139,433,351,414,374,383,221,401,53,4,434,218 362,196,302,138,358,63,449,65,441,376,256,275,365,427,455,496,410,283,299,464,486,72 108,179,145,102,270,92,306,407,235,413,231,99,323,249,479,18,294,184,394,461,439,354 286,391,489,280,368,250,200,245,193,304,278,271,303,498,152,268,96,154,484,84,300,163 385,253,473,75,381,37,277,113,436,467,16,440,105,42,509,66,463,320,203,11,402,313,166 27,206,488,315,240,49,307,500,78,211,285,390,293,428,70,201,350,153,56,505,14,477,344 141,57,438,511,118,233,370,287,130,21,194,290,499,103,164,352,421,276,395,131,26,230 109,321,460,462,195,124,375,20,269,406,137,279,82,5,326,333,28,111,312,337,332,178,169 425,416,122,117,494,443,173,254,150,215,485,418,429,162,219,466,238,501,424,181,32,359 502,183,227,432,261,80,19,12,308,301,114,121,357,360,504,431,127,46,25,90,430,491,405 225,68,310,220,444,317,91,318,2,79,373,503,223,182,408,156,248,284,382,247,135,327,305 81,125,369,347,442,386,170,148,289,87,52,324,241,403,468,176,222,480,363,83,142,420 267,39,265,367,77,45,507,393,143,251,144,190,372,348,482,472,54,10,31,419,478,100,325 237,180,282,94,426,319,205,132,356,469,51,483,335,115,217,411,291,59,133,452,288,244 22,60,204,316,458,487,89,226,346,36,398,273,189,242,260,343,167,177,339,292,192,314 404,110,198,475,353,191,3,73,175,445,349,296,474,236,24

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X^5	0,1,32,51,34,295,83,324,98,227,466,321,53,432,468,67,38,243,142,73,397,92,372,183,147 198,427,492,333,28,36,359,226,229,159,138,328,75,356,117,56,191,469,320,487,100,91 458,249,42,12,205,218,269,126,443,244,167,401,464,162,501,406,211,174,158,78,108,377 77,456,238,84,228,292,390,502,66,215,369,307,471,347,429,365,141,340,166,286,378,230 144,309,85,156,494,479,489,354,326,384,178,364,76,413,299,176,20,439,5,203,363,127 157,74,186,425,79,461,57,234,136,335,319,329,47,189,457,362,200,345,233,388,290,486 338,71,357,484,212,220,250,302,282,213,419,110,266,434,448,344,312,303,217,4,480,61 207,327,423,323,231,301,155,37,133,26,168,214,498,296,30,453,485,106,88,193,433,39 325,46,90,153,255,387,115,245,279,281,493,62,216,45,446,222,351,310,417,404,273,408 139,251,506,246,225,452,449,41,366,82,263,187,248,145,192,331,140,160,373,428,367 22,199,407,2,313,190,260,405,507,376,154,399,420,163,124,109,275,272,430,239,392,219 180,241,195,148,116,437,450,17,283,478,252,43,54,414,460,118,101,201,462,368,280,48 370,72,318,274,261,315,164,472,470,184,382,262,93,55,349,161,447,81,242,10,65,171,125 467,474,102,422,268,80,232,491,197,476,224,69,111,35,27,210,424,509,149,128,254,510 402,403,105,300,196,436,330,346,438,308,173,14,389,146,15,505,374,130,411,40,291,337 332,426,421,29,336,431,240,50,123,465,394,380,177,350,129,294,495,341,398,236,371,395 6,194,89,500,381,482,253,277,24,441,418,287,278,504,179,23,334,95,16,481,444,33,490 94,391,499,316,477,256,386,393,107,114,285,18,165,440,483,360,410,259,265,134,289,188 137,86,488,293,415,68,175,358,63,96,206,131,348,7,508,181,258,271,182,169,21,284,496 235,475,342,511,352,185,304,204,343,52,237,264,451,170,375,455,8,58,99,150,221,209 396,44,355,361,339,64,104,3,317,379,87,400,298,297,385,143,305,103,459,473,311,120 132,314,208,202,306,503,409,454,442,353,11,257,121,122,70,270,288,152,416,445,151,59 135,223,113,172,276,25,435,247,31,267,497,412,112,49,463,97,9,13,119,383,19,322,60
X^9	0,1,17,274,257,448,186,377,409,481,302,84,335,503,82,488,197,125,270,180,343,303,310,76 277,468,120,443,336,337,407,148,243,228,414,139,358,433,417,116,141,227,326,42,207,97 174,258,205,99,122,470,459,421,214,442,250,45,235,318,43,60,400,133,436,94,211,59,178 408,127,87,445,46,124,237,364,63,263,342,25,330,164,245,396,31,155,10,89,371,66,106,27 497,426,322,123,391,96,158,489,213,88,102,405,16,296,431,464,149,199,128,301,104,236 171,44,425,71,192,138,438,493,467,92,416,145,111,41,75,412,252,389,295,410,58,455,476 212,461,444,359,261,220,176,107,223,262,399,404,74,339,279,437,478,126,511,413,156,508 393,509,320,54,177,5,466,100,384,397,495,224,367,418,170,357,491,294,248,311,64,77,249 502,427,287,30,424,471,419,456,190,504,113,315,432,83,282,314,369,390,118,483,273,506 202,53,7,9,313,272,290,305,193,386,368,62,375,385,458,209,344,196,79,356,251,219,70,91 260,78,19,253,283,484,256,21,307,166,130,110,328,11,47,450,36,13,233,446,225,381,288 453,90,172,49,168,507,217,136,144,259,331,474,281,51,462,486,246,284,395,355,345,179 242,26,498,216,499,475,161,306,428,317,477,142,378,299,485,160,232,175,329,204,494,361 435,143,24,38,200,308,201,55,239,275,56,198,208,492,429,403,240,373,129,6,280,93,195 132,469,365,210,360,490,402,327,61,500,309,85,406,28,29,23,276,332,333,401,146,480,289 151,340,292,348,351,37,95,231,398,52,420,266,479,115,271,353,222,434,98,181,191,362 286,265,105,380,454,465,103,370,510,297,501,32,73,39,334,34,422,264,267,167,135,439 338,352,18,482,109,159,321,8,50,121,3,394,218,81,298,163,293,430,300,101,137,194,165 341,12,254,372,68,119,69,153,447,48,20,152,382,411,383,184,487,183,234,366,241,203,86 463,80,188,33,349,2,388,473,423,65,114,150,226,452,157,185,415,324,316,229,269,278,4 285,457,363,460,108,140,238,291,67,346,312,35,323,347,505,392,40,325,350,154,387,147 72,230,319,189,112,354,173,187,182,206,449,268,440,117,451,221,169,14,376,131,247 134,496,22,162,441,15,379,374,472,215,57,244,304,255

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X^{65}	0,1,164,245,3,394,230,319,492,208,434,222,287,427,256,484,5,177,233,13,143,435,34,334 298,163,316,229,336,337,263,342,37,351,51,281,368,386,295,389,199,128,43,60,354,173 463,80,304,255,366,241,236,171,243,228,273,483,437,279,61,327,216,498,15,441,382,152 467,493,227,141,315,432,176,107,23,276,477,142,401,146,168,507,196,79,444,359,102,88 421,459,451,117,65,423,367,418,172,49,485,160,103,370,341,165,108,460,303,343,87,127 481,409,106,66,482,18,40,392,280,93,105,380,491,294,219,70,111,145,97,207,496,134 447,153,85,406,161,306,314,369,399,404,385,458,455,476,151,340,144,259,376,14,452 226,158,96,99,205,329,204,501,32,384,397,381,288,125,197,59,211,68,372,67,291,439 135,323,35,503,335,322,426,64,77,78,19,240,373,191,362,321,8,154,387,257,448,155,10 419,471,130,166,275,239,115,479,308,200,167,267,509,393,47,11,277,468,124,237,300 101,4,285,290,272,75,41,52,398,284,246,206,449,349,2,296,431,250,45,71,192,358,433 472,215,184,487,360,210,179,345,7,53,413,511,17,274,25,330,218,81,147,72,403,429 353,271,424,30,283,253,100,466,36,450,38,24,39,73,293,430,415,324,407,148,364,63 348,292,486,462,193,305,58,410,464,149,400,133,189,112,188,33,57,244,203,86,301,104 414,139,118,390,126,478,402,490,475,499,162,22,383,411,438,138,42,326,504,113,223 262,28,29,378,299,332,333,217,136,209,344,261,220,213,489,442,214,440,268,150,114 170,357,453,90,232,175,454,465,254,12,363,457,76,310,408,178,84,302,371,89,159,109 505,347,195,132,286,265,248,311,356,251,416,92,258,174,247,131,20,48,500,309,428 317,83,282,74,339,62,375,212,461,480,289,331,474,169,221,185,157,391,123,470,122 494,361,510,297,495,224,446,225,180,270,94,436,69,119,238,140,352,338,312,346,488 82,497,27,249,502,91,260,129,6,98,181,50,121,325,350,186,377,396,31,190,456,307,21 198,56,266,420,55,201,264,422,54,320,328,110,120,443,445,46,137,194,269,278,313,9 252,412,231,95,355,395,187,182,388,473,405,16,235,318,44,425,417,116,379,374,183 234,365,469,26,242,202,506,508,156
$X^3 \oplus X^5$	0,0,40,60,98,370,43,303,115,187,363,439,484,60,157,337,174,362,95,399,103,179,247,55 108,352,429,181,336,332,240,248,128,132,49,33,499,231,291,35,287,467,158,70,409,69,121 433,306,242,346,142,234,58,227,39,28,276,324,88,49,41,264,260,431,342,343,442,43,450 178,335,430,415,102,67,479,254,118,323,79,370,110,327,352,333,288,281,255,266,494,15 37,192,341,420,496,269,401,376,357,136,357,156,29,40,332,365,381,88,77,380,268,53,436 153,306,283,491,470,80,417,472,61,411,378,114,135,294,52,366,104,264,266,289,311,130 344,506,52,345,403,64,158,370,420,483,289,503,49,263,213,103,121,454,460,279,25,215 461,246,480,39,293,457,463,377,363,190,352,351,149,116,186,500,302,446,364,438,368,42 488,67,405,327,349,127,113,294,44,127,353,341,190,461,50,157,102,100,139,131,416,299 28,190,141,375,336,79,96,14,53,300,19,268,39,296,463,89,170,446,73,174,333,90,437,91 416,131,124,227,8,96,327,337,98,332,379,28,63,92,119,132,187,46,277,151,440,215,52 319,456,336,163,217,318,71,42,31,102,477,160,484,141,346,511,50,131,309,128,60,413 191,278,62,387,398,311,366,451,275,114,162,471,471,422,7,98,291,330,482,415,424,209 264,101,466,371,35,150,172,285,316,153,199,362,223,358,231,90,158,55,135,482,431,222 338,295,27,122,250,110,114,242,134,258,111,511,405,457,45,101,28,336,453,157,332,28 285,89,411,475,427,511,146,266,499,127,432,312,176,44,321,465,336,468,44,428,92,456 450,410,227,175,346,18,26,326,491,191,291,99,45,105,132,212,473,69,33,425,490,358,115 235,23,360,47,324,449,430,408,483,221,362,469,118,254,89,407,292,21,174,244,91,360 195,488,87,110,29,447,472,230,389,342,33,35,344,130,493,228,143,36,91,261,182,148,307 55,148,455,368,317,386,325,494,337,254,328,243,170,221,482,385,307,84,26,361,374,240 414,12,326,464,463,333,462,128,22,332,11,85,434,504,58,120,11,93,161,499,241,439,51 185,306,428,349,199,61,435,413,31,492,122,188,46,172,42,201,387,392,214,29,71,317,371 461,395,357,311,71,273,142,460,40,166,432,298,87,457,430,36

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$X^3 \oplus X^9$	0,0,25,285,321,405,194,274,392,441,407,162,158,123,283,250,77,484,479,370,189,448,181 204,490,114,126,226,333,1,323,267,145,133,304,32,477,285,486,34,426,399,13,300,177,64 140,377,262,187,300,405,507,146,75,38,18,414,62,182,184,480,270,338,181,406,458,493,480 23,261,502,71,341,318,296,325,131,422,356,357,495,401,31,385,479,239,437,440,259,330 245,267,356,99,264,84,355,147,160,268,239,81,182,21,275,212,214,282,200,65,151,94,192 274,136,183,253,97,303,48,159,378,209,142,245,94,289,101,183,427,125,265,271,349,95 258,481,202,301,57,14,107,344,279,108,338,301,458,101,21,190,95,277,28,82,213,331,268 406,60,250,74,392,349,335,177,423,488,287,408,107,222,253,308,19,148,507,361,258,68 511,35,156,367,49,148,206,488,98,393,263,128,369,40,221,504,221,202,235,157,349,51,247 434,166,134,150,111,55,332,16,166,42,31,407,93,52,376,21,195,126,124,453,169,332,441 344,476,237,342,355,7,467,273,449,293,37,425,429,412,464,263,79,344,448,89,453,285,352 384,249,398,295,137,292,217,79,266,152,296,362,353,39,347,508,142,301,253,142,178,453 322,125,282,289,258,233,192,47,239,481,177,187,248,290,316,482,187,57,208,342,327,273 182,484,394,313,487,80,33,70,470,181,250,465,282,309,183,328,205,54,484,254,2,28,510 48,386,328,27,430,174,31,510,159,209,180,227,359,80,208,337,1,120,44,29,1,291,59,73 129,237,289,202,231,498,219,201,48,107,406,9,424,260,417,481,148,374,263,66,466,329 477,509,185,364,300,469,477,339,95,396,336,144,328,433,392,305,12,447,338,165,332,274 342,278,86,206,94,336,196,127,10,125,268,500,341,108,457,438,91,57,208,475,226,462,499 244,296,381,421,206,454,221,209,440,488,4,336,105,237,79,463,102,7,476,185,480,341,448 113,198,319,241,12,166,395,267,290,311,255,262,458,256,28,171,179,384,231,226,129,72 507,176,7,407,193,243,161,8,394,246,112,185,119,80,410,192,218,435,173,129,126,110,405 175,132,474,245,346,41,384,503,159,312,479,380,510,188,290,356,108,506,298,440,441,355 232,310,461,451,262,12,306,473,101,394,273,302,476,231
$X^3 \oplus X^5 \oplus X^{17}$	0,1,416,371,215,363,325,43,333,39,137,305,410,333,108,105,428,43,87,258,93,103,404,380 188,80,291,285,333,28,224,355,355,112,107,426,469,379,239,147,236,148,384,298,90,415 260,19,63,170,364,299,431,135,206,308,493,19,218,502,125,62,376,489,29,55,209,41,112 487,142,459,36,357,140,287,73,181,211,253,433,29,294,88,250,235,95,156,469,274,294,307 158,484,95,503,175,407,203,289,163,38,245,162,84,7,84,213,88,438,106,342,499,333,460 416,217,474,212,261,341,128,270,9,127,23,22,172,96,365,428,115,137,57,375,277,337,311 249,77,440,99,34,299,198,77,337,264,265,63,172,328,426,74,89,363,101,56,420,299,181 170,465,284,61,415,363,27,326,50,70,480,462,263,252,231,227,378,476,151,333,361,64,182 333,447,22,54,227,428,394,23,119,337,215,291,36,191,182,255,50,127,246,105,97,401,151 437,209,113,42,88,420,185,365,162,201,258,86,335,444,458,273,437,371,327,379,413,321 200,379,32,500,171,408,21,454,292,408,424,293,151,374,22,49,62,80,141,511,294,456,451 235,399,238,344,135,103,47,29,413,192,263,136,65,458,141,468,347,365,421,321,473,191 298,158,485,318,292,301,322,335,469,266,382,206,475,185,375,133,375,87,12,67,62,163 371,490,279,348,8,300,94,424,217,173,130,36,132,333,237,502,128,415,191,370,221,127,208 160,323,392,135,158,227,405,277,177,241,337,337,35,337,332,195,12,29,336,386,29,155,107 310,276,498,468,9,253,372,239,189,500,354,187,14,261,163,199,509,331,18,160,282,378,467 220,233,308,204,147,507,374,43,457,302,30,481,213,178,340,262,399,103,60,364,128,424 150,72,25,190,61,470,337,374,291,242,456,96,392,312,338,423,287,314,237,407,146,479,222 292,247,477,353,276,378,298,468,326,362,111,300,49,416,338,199,346,29,23,63,45,215,398 246,441,19,493,296,488,511,427,440,504,313,456,102,425,213,162,357,266,31,316,326,166 14,364,448,160,222,242,483,268,207,246,183,261,406,78,434,399,161,357,79,242,266,477 330,120,61,309,480,53,50,202,418,504,66,57,391,349,49,454,453,144,65,401,194,202,331 328,422,33,29,192,248,511,277,25,412,276,67